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Technical Report No. 74

Reprocessing and Reworking of Biologicals

Revised 2026

PDA Parenteral Drug Association

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Revised 2026

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1.0 Introduction

Reprocessing and reworking are tools that biological manufacturers can use when a manufacturing process step has operated outside of its validated operating ranges. For reprocessing, rather than proceeding with the next manufacturing process step, the product is rerun through the previous manufacturing process step operated within the acceptable operating ranges. For reworking, the product is subjected to one or more processing steps that are different from the established manufacturing process to obtain acceptable product. In biological manufacturing, however, reprocessing, and especially reworking, should be considered only in exceptional circumstances. Reworking can be considered during events such as critical equipment failures, out of specifications of critical quality attributes of drug substance or drug product, and/or deviation from in-process controls that impact quality of the drug substance or drug product (further conditions to consider are stated in **Section 5.0**).

Reprocessing and reworking are deviations that are managed within the corrective and preventive actions (CAPA) of the manufacturer's quality management system. An essential prerequisite for the acceptance of a reprocessing or reworking step is a clear indication of the root cause. Due to the size and complexity of biological products, including the limitations in testing methods (especially measurement of functional biological activity, i.e., potency), reprocessing, and especially reworking, are limited for biologicals. Thus, they should be entered into with caution so as not to impact the product's critical quality attributes (CQA), and the costs and risks should be carefully evaluated and documented.

Effective reprocessing and reworking strategies should integrate supply chain resilience considerations, including forecasting, inventory buffers, and manufacturing scheduling. Combining quality risk management assessments (e.g., a FMEA assessment) with supply risk scoring can support balanced decisions that protect both compliance and patient access.

Reprocessing and reworking also require regulatory authority approval before the manufactured product can be released for human use. In limited cases, regulatory authority approval for reprocessing can be acquired ahead of time by presenting a comprehensive change management plan (proactive approach detailed in **Section 3.6**). This plan should explain the reason for reprocessing, such as refiltration or reconcentration due to technical failures or mechanical breakdown of a chromatography column ***(1)***. It must also specify the frequency and nature of the reprocessing procedures for potential events, as well as detail the steps for validation and ensuring product comparability.

1.1 Purpose

PDA *Technical Report No. 74 (Revised 2026): Reprocessing and Reworking of Biologicals* (TR 74) presents a strategy based on the experiences and practices of a PDA Technical Report Team representing a cross-section of industry professionals. The intent is to provide guidance on the design, development, controls,

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procedures, validation, regulatory submission, and implementation of reprocessing and reworking procedures for the manufacturing of biologicals.

The information discussed represents what the authors consider a balance of scientific, regulatory, and business considerations for the reprocessing of biologicals in particular but can also be applied to the reworking of biologicals in general. It is highly recommended that manufacturers consult with the appropriate regulatory authority for agreement on the strategies employed for any biological reprocessing or reworking operation.

1.2 Scope

The previous version of TR 74 included only such biopharmaceutical products as recombinant proteins and polypeptides. Because the principles of reprocessing and reworking are applicable across a wider array of manufacturing processes, the scope has been expanded to include guidance on biological products. In addition to recombinant proteins and monoclonal antibodies (both as innovators/novel and biosimilars), biological products ***(2-4)*** encompass a wide range of other products such as vaccines and blood-derived proteins.

The quality of biologicals is defined by the chosen production and manufacturing process. Minor changes in the process can affect the quality of the drug product; therefore, the development of the manufacturing process is of paramount importance for biologicals, including the implementation of any reprocessing or reworking. Incorporating the topic of reworking further expands the scope of this technical report, but because the reworking of biologicals is far more challenging than reprocessing, the authors emphasize that extra caution should be taken.

The guidance in this technical report is general in nature, and the case studies illustrate how the general principles may be applied. No attempt is made to encompass all types of unit operations for which reprocessing or reworking could be considered.

This technical report is not intended to establish mandatory standards but serves as a single-source overview that complements existing guidance documents listed in **Section 8.0**.

The principles discussed in TR 74 are applicable to both clinical and commercial manufacturing processes, but the authors recognize that reprocessing and reworking during the clinical manufacturing of biologicals is often more challenging than during commercial manufacturing. This is primarily because the understanding of the product and the process is very limited, so extra caution is emphasized.

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2.0 Glossary

Definitions have been provided to help clarify the concepts discussed in this document. While some definitions used vary among companies, geographic location, etc., the definitions described below are consistently used within this technical report. Where a definition is based on another published source, the source is cited.

Acceptance Criteria
Numerical limits, ranges, or other suitable measures for acceptance of the results of analytical procedures which the drug substance or drug product or materials at other stages of their manufacture should meet ***(2)***.

Action Limit
An internal (in-house) value used to assess the consistency of the process at less critical steps ***(2)***.

Active Pharmaceutical Ingredient (API) (or Drug Substance)
Any substance or mixture of substances intended to be used in the manufacture of a drug product and that, when used in the production of a drug, becomes an active ingredient of the drug product. Such substances are intended to furnish pharmacological activity or other direct effect in the diagnosis, cure, mitigation, treatment or prevention of disease, or to affect the structure and function of the body ***(3)***.

Contaminants
Any adventitiously introduced materials (e.g., chemical, biochemical, or microbial species) not intended to be part of the manufacturing process of the drug substance or drug product ***(2)***.

Critical Process Parameter (CPP)
A process parameter whose variability has an impact on critical quality attribute and therefore should be monitored or controlled to ensure the process produces the desired quality ***(4)***.

Critical Quality Attribute (CQA)
A physical, chemical, biological, or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure the desired product quality. CQAs are generally associated with the drug substance, intermediates, excipients, intermediates (in-process materials) and drug product ***(4)***.

Design of Experiments (DOE)
An applied statistical method that deals with planning, conducting, analyzing, and interpreting controlled tests to evaluate the factors that control a process parameter output.

Design Space
The multidimensional combination and interaction of input variables (e.g., material attributes) and process parameters that have been demonstrated to provide assurance of quality. Working within the design space is not considered as a change. Movement out of the design space is considered to be a change and would normally initiate a regulatory post approval change process. Design space is proposed by the applicant and is subject to regulatory assessment and approval ***(4)***.

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Drug Product (Dosage Form; Finished Product)
A pharmaceutical product type that contains a drug substance, generally, in association with excipients ***(2)***.

Drug Substance
The material which is subsequently formulated with excipients to produce the drug product. It can be composed of the desired product, product related substances, and product- and process-related impurities. It may also contain excipients including other components such as buffers ***(2)***.

Failure Mode and Effects Analysis (FMEA)
FMEA (see IEC 60812) provides for an evaluation of potential failure modes for processes and their likely effect on outcomes and/or product performance. Once failure modes are established, risk reduction can be used to eliminate, contain, reduce or control the potential failures. FMEA relies on product and process understanding. FMEA methodically breaks down the analysis of complex processes into manageable steps. It is a powerful tool for summarizing the important modes of failure, factors causing these failures and the likely effects of these failures ***(5, 6)***.

Impurity
Any component present in the drug substance or drug product which is not the desired product, a product-related substance, or excipient including buffer components. It may be either process- or product-related ***(2)***.

Note: Impurities are to be distinguished from contaminants, which are defined above.

In-process Control (or Process Control)
Checks performed during production to monitor and, if appropriate, to adjust the process and/or to ensure that the intermediate or API conforms to its specifications ***(3)***.

Intermediate Drug Substance
A material produced during steps of the processing of an API that undergoes further molecular change or purification before it becomes an API ***(3)***.

Note: The term "intermediate" in this document applies to both intermediates of the drug substance (API) and intermediates of the drug product, unless otherwise specified.

Drug Product
A material that is the result of a drug substance having undergone at least one processing step in the presence of any other substance(s) (used in the manufacture of the drug product whether it appears in the finished dosage form or not) which must undergo further processing step(s) to become the finished dosage form ***(7)***.

Prior Approval
Regulatory requirement to approve a change that has a substantial potential to have an adverse effect on the identity, strength, quality, purity or potency of the drug product as these factors may relate to the safety or efficacy of the drug product ***(6, 8, 9)***.

> ***For Clinical/Investigation Products***
> European Medicines Agency (EMA) requires prior approval of substantial modifications; U.S. Food and Drug Administration (FDA) requires submission of information amendments.
>
> ***For Commercial/Marketed Products***
> European Medicines Agency (EMA) requires prior approval of Type II variations; FDA requires a prior approval supplement (PAS) to the Biological License Application (BLA).

Process Validation
The collection and evaluation of data, from the process design stage through commercial production, which establishes scientific evidence that a process is capable of consistently delivering quality products ***(10)***, or the documented evidence that the process, operated within established parameters, can perform effectively and reproducibly to produce a

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medicinal product meeting its predetermined specifications and quality attributes ***(11)***.

Prospective Process Validation

Validation conducted prior to the distribution of either a new product or a product made under a revised manufacturing process where the revisions may affect the product's characteristics ***(3)***.

Reprocessing

Introducing an intermediate or API, including one that does not conform to standards or specifications, back into the process and repeating a crystallization step or other appropriate chemical or physical manipulation steps (e.g., distillation, filtration, chromatography, milling) that are part of the established manufacturing process ***(3)***.

Note: Reprocessing also applies to drug product intermediates and drug products. In biological manufacturing, these situations are usually restricted to some refiltration, re-concentration steps upon technical failure of equipment or mechanical breakdown of a chromatography column.

Reworking

Subjecting an intermediate or API that does not conform to standards or specifications to one or more processing steps that are different from the established manufacturing process to obtain acceptable quality intermediate or API (e.g., recrystallizing with a different solvent) ***(3)***.

Note: Under some extraordinary situations, a biological bulk drug substance with host cell protein levels slightly above specification limits may be reworked via an additional, validated chromatography step under a pre-approved protocol.

Specification

A list of tests, references to analytical procedures, and appropriate acceptance criteria, which are numerical limits, ranges, or other criteria for the tests described. It establishes the set of criteria to which a drug substance, drug product, or materials at other stages of its manufacture should conform to be considered acceptable for its intended use. "Conformance to specification" means that the drug substance and drug product, when tested according to the listed analytical procedures, will meet the acceptance criteria. Specifications are critical quality standards that are proposed and justified by the manufacturer and approved by regulatory authorities as conditions of approval ***(2)***.

Unit Operation (or Process Step)

A discrete step or manipulation in a manufacturing process where process and operating parameters are defined to achieve a specific process objective. (e.g., chromatography, depth filtration, virus filtration, ultra-filtration/diafiltration, bioburden reduction filtration, freeze/thaw formulation and filling)

Validation

A documented program that provides a high degree of assurance that a specific process, method, or system will consistently produce a result meeting predetermined acceptance criteria ***(3)***.

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2.1 Abbreviations

AEX	Anion Exchange Chromatography
API	Active Pharmaceutical Ingredient
BLA	Biological License Application
CCS	Contamination Control Strategy
CDMO	Contract Drug Manufacturing Organization
CMC	Chemistry, Manufacturing, and Control
CP	Comparability Protocol
CPP	Critical Process Parameter
CQA	Critical Quality Attribute
CTD	Common Technical Document
DOE	Design of Experiments
EC	Established Conditions
FMEA	Failure Mode and Effects Analysis
HCP	Host Cell-derived Proteins
ICH	International Council for Harmonisation
IEX	Ion Exchange Chromatography
IPC	In-process Control
Mab	Monoclonal Antibody
MAH	Manufacturing Authorization Holder
MSAT	Manufacturing Science and Technology
OOS	Out-of-Specification
PACMP	Post-Approval Change Management Protocol
PAS	Prior Approval Supplement
PQS	Pharmaceutical Quality System
QRM	Quality Risk Management

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3.0 Reprocessing Considerations

Reprocessing ***(3, 9, 11-15)***, or repetition of an existing biopharmaceutical unit operation, requires a well-designed process, supportive data, and validation to augment the existing manufacturing process. Importantly, the supportive data must provide the manufacturer with a high degree of confidence (assurance) that the drug substance or drug product produced by reprocessing meets the predefined quality standards for the drug substance or drug product and is comparable to a nonreprocessed batch. The reprocessed and reworked biological drug substances and sterile drug products should adhere to all CQAs associated (including bioburden and endotoxins) to normal batch disposition and release. Enhanced testing protocols are often required to assess the quality attributes of the reprocessed material, ensuring that it remains consistent with the established manufacturing process.

Reprocessing in pharmaceutical manufacturing can occur under various scenarios that can be broadly categorized as proactive and reactive approaches.

In GMP, a risk-based approach ***(6),*** to reprocessing is incorporated by systematically assessing and managing potential risks associated with reactive reprocessing activities. This begins with identifying and evaluating the nature of deviations and their potential impact on product quality and patient safety. By prioritizing areas of greatest concern, manufacturers can focus resources and controls where they are most needed. Decisions regarding reprocessing are made based on these risk assessments, ensuring that only safe and effective practices are implemented.

For any event that requires reprocessing, the manufacturer must perform a formal root-cause investigation of the unit operation excursion if the excursion if not within the design space. Additional development studies may be required to supplement the root-cause investigation. Consideration of any proposed reprocessing by regulators will likely be based on the root-cause analysis and the impact of reprocessing on product quality.

3.1 Proactive Approach

In a proactive scenario, the potential need for reprocessing is identified in advance through predetermined criteria. The execution of the reprocessing activities occurs through a predefined concurrent validation protocol with clear acceptance criteria or through previously validated procedures may be reviewed and approved by applicable regulatory agencies. This approach includes:

1. **Predefined Criteria:** Manufacturers outline specific conditions or triggers that would necessitate reprocessing, allowing for a structured response to an event that may require reprocessing. Post reprocessing, there should be sufficient data to support a conclusion that any given reprocessed batch of a drug substance or drug product is uniform and meets the defined acceptance criteria.

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2. **Validated Procedures:** Reprocessing is executed according to a validated procedure that has undergone detailed review by the manufacturer's quality unit. This concurrent validation ensures that the process is capable of consistently producing a product that meets established quality standards. The validation protocol should include the scope of the manufacturing process that is being reprocessed, number of batches required, tests to be performed, and acceptance criteria. The proposed acceptance criteria for all tests should be appropriately justified. The concurrent validation approach should be described within the Validation Master Plan. An example of reprocessing that might be specifically covered by a Biological License Application (BLA) would be repeating viral filtration following a failed filter-integrity test. Because viral filters occasionally fail, manufacturers may be approved for refiltration of the protein drug substance or drug product (following successful validation). The approach to validation should be consistent with a lifecycle approach to validation, as discussed in the U.S. Food and Drug Administration *Guidance for Industry: Process Validation: General Principles and Practices* and PDA *Technical Report No. 60 (Revised 2026) Process Validation: A Lifecycle Approach* ***(10)***.

   When reprocessing has been prospectively validated, it is appropriate to provide the full validation protocol and report in a regulatory submission and to follow the validated process where indicated. When reprocessing is anticipated to undergo concurrent validation, the protocol—inclusive of initial starting conditions and appropriate acceptance criteria to determine whether successful completion of a reprocessing step has been achieved—should be provided.

3. **Regulatory Guidance derived from Established Protocols:** An essential prerequisite for the acceptance of a reprocessing step is a clear identification of the root cause. For biological products, these situations are usually restricted to some refiltration, reconcentration steps upon technical failure of equipment, or mechanical breakdown of a chromatography column. These steps should be appropriately described, validated, and included in the regulatory submission. Such documentation should include the description of conditions for which reprocessing could be applied and a demonstration that the reprocessing step(s) does not impact the quality of the drug substance and/or drug product. The sequence of steps detailed in **Section 3.6** provides a roadmap for implementing proactive reprocessing, ensuring that all necessary evaluations and approvals are addressed before action is taken.

3.2 Reactive Approach

Reactive reprocessing occurs in response to unforeseen circumstances during the manufacturing process. The same categories of data and decisions required for a proactive reprocessing approach apply to this approach, except that less time and information may be available for performing assessments. As a result, reactive reprocessing is better suited for steps that are simple (such as microbial or virus filtration after failure of filter-integrity testing) and have a low likelihood of affecting product quality. The sequence of steps discussed in **Section 3.6** may be used to guide the approach to reactive reprocessing, starting with the design of a reprocessing unit operation (see **Section 3.6, Step 6**).

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Two distinct reactive reprocessing scenarios are described as follows:

1. **Nonprospectively Validated Reprocessing at an In-Process Hold Point**
    a. In this scenario, a need arises for a nonprospectively validated reprocessing unit operation while a product is held at an in-process stage. This may involve scenarios such as freezing the product for an extended period.
    b. During the hold period, manufacturers can develop and execute a plan to validate the reprocessing step. This provides assurance that the product meets all the required quality attributes subsequent to the product concurrent release.
2. **Immediate Reprocessing During a Manufacturing Run**
    a. This scenario occurs when a reprocessing need emerges unexpectedly during a manufacturing run. In such cases, the reprocessing step may need to be implemented before its validation.
    b. If a rapid design and implementation of the reprocessing option is feasible, it may be executed "at risk." This means that the reprocessing can occur with the understanding that supportive data must be generated and reviewed to ensure product quality before the batch is released to the market.

Of the two reactive reprocessing scenarios described above, the second scenario (#2) carries the highest degree of risk and is likely to be implemented only with relatively simple unit operations that affect a limited number of CQAs, such as dead-end filtration for drug substance or drug product. Release of the reprocessed batch using a reactive scenario should only be performed upon receiving regulatory approval. (See **Section 4.0** for strategies on regulatory submissions).

A proactive process design approach that accounts for reprocessing could apply to more complex (such as ultrafiltration, diafiltration, concentration, chromatography, viral removal steps, and polishing) steps with significant benefit to the manufacturer.

In principle, almost any unit operation may be considered for potential reprocessing as long as it is demonstrated to consistently deliver a drug substance or drug product that meets the same specified quality commitments of the standard manufacturing process. Reprocessing simple unit operations is relatively well accepted and understood by regulators and the industry. Reprocessing a sterile drug substance and reprocessing and reworking of sterile-biological batches requires critical attention to maintaining sterility, endotoxin, and particulate controls. Reprocessing of nonsterile drug substances manufactured in low bioburden facilities also requires special attention to be paid to microbial and particulate controls.

3.3 Reprocessing Frequency

A manufacturing process that requires frequent reprocessing is considered to be poorly controlled and not in a validated state. While there is no formal definition of “frequent,” it would be reasonable to question the state-of-control for a unit operation that requires reprocessing as often as 10% of the time ***(16)***. (Ten percent is given here as an illustrative example and is not intended to be a prescriptive limit.)

It is important to distinguish between two types of reprocessing frequency:

- Number of reprocessed unit operations within a batch
- Number of batches that require reprocessing over a defined time period

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Given that the Annual Product Review/Product Quality Review provides a one-year performance summary, it may be appropriate to evaluate reprocessing trends within that annual timeframe. Under circumstances where reprocessing becomes recurring or trends upward, a reassessment of the process design should be strongly considered. If reprocessing is carried out frequently or reaches a specified percentage of manufactured batches, it must be considered part of the normal process, described as such in the application, and receives approval.

3.3.1 Reprocessing Frequency per Batch

The frequency of reprocessing for the same batch at commercial-scale reprocessing should be restricted to a maximum number of reprocessing events according to process development data for small-scale reprocessing.

Validation of the reprocessing step should be supported by a scientifically justified number of demonstrations runs. While no fixed minimum is mandated by regulatory agencies, performing at least two independent runs is common practice to demonstrate reproducibility. The exact number should be based on process complexity, risk assessment, and historical data to ensure the process consistently achieves predefined acceptance criteria.

The reprocessing frequency is established by the manufacturer based on appropriate studies at either small scale or same scale. Small-scale validation studies are commonly used to establish frequency for a reprocessing event for a particular unit operation. These studies may be conducted before implementation of a reprocessing approach at manufacturing scale. In a reprocessing event, product quality can change after repeated run exposures during filtration or chromatography processing steps and may impact the stability of long-term storage. There is no rule-of-thumb for frequency determination; the manufacturer must establish it for each project based on its impact on quality attributes. For example, a simple dead-end filtration, typically a maximum of three reprocessings, can be included in a small-scale reprocessing study. Some protein molecules may not be stable for a second reprocessing; in that case, maximum reprocessing may be restricted to only one reprocessing event.

These studies are used to obtain data on product quality attributes (e.g., aggregation) resulting from exposure to physical/shear stress during repeated unit operations. Small-scale validation-study design should be predictive of the commercial process. This will ensure that the data generated is relevant to large-scale. Small-scale reprocessing studies must use the same protein solution for the repeated number of runs and the samples to be withdrawn after each run. Product quality attributes of each run should be compared against the initial and intermediate feed-material to understand the impact of multiple exposures and to determine the frequency for the large-scale reprocessing event. All reprocessing events, including repeated runs, need to be completed within the allowed chemical and microbial hold time for intermediate pools or scientifically justified if there is any excursion from validated hold time.

When developing a proactive reprocessing step, a review of the development data and history may identify process steps with inherent variability that may, on rare occasions, lead to a need for reprocessing (e.g., failing in-process control (IPC) assay results, filter integrity failure, difficult separation step). The variability must be scientifically understood.

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3.4 Continuation vs. Reprocessing (Repetition) of a Unit Operation

Most unit operations include some sort of in-process control measurement to determine whether the step meets the objective (e.g., a concentration, pH). An adjustment to bring a product within acceptable limits is not considered reprocessing and, instead, should be designed as a process "continuation." Procedures should account for continuation to perform an adjustment within defined limits. Without appropriate instructions, continuations in processing may be erroneously considered as reprocessing and lead to procedural deviations.

For example, during a step in the ultrafiltration-diafiltration operation, concentration is measured at different volumes to check for the desired target concentration. If the concentration does not meet the target specification, the process can be continued until the target specification is met, and it is considered a continuation of the process, not reprocessing. Similarly, in the titration process, the addition of titrant to achieve the desired pH or conductivity set point may be considered a continuation of the process. Likewise, adding a buffer dilution to achieve the target product concentration is considered a continuation of the process, not a reprocessing activity.

In the case of drug product manufacturing, when large-volume containers are thawed and only a partial amount is used for filling, the remaining formulated drug substance may need to be refrozen for subsequent use. This practice may be considered reprocessing if not included in an approved application. Supporting freeze-thaw stability data from reduced- or-at-scale studies can help justify the impact on drug product quality.

3.5 Quality Concerns During the Reprocessing of Biologicals

Protein aggregation is a key concern during manufacturing and quality assessment, particularly in the context of reprocessing. Most therapeutic proteins are only marginally stable in solutions and are susceptible to chemical instabilities (e.g., deamidation, oxidation, hydrolysis) as well as physical instabilities (e.g., denaturation and noncovalent aggregation).

During reprocessing, proteins may be exposed to various stress conditions. Process controls play a critical role in mitigating the effects of destabilizing factors, such as pH, buffer components, and denaturants, that can induce protein unfolding and subsequent aggregation.

Several stress factors during reprocessing can contribute to protein aggregation and instability. Key considerations include:

- **Increased Viscosity During Reprocessing:** Higher protein concentration can increase viscosity, affecting subcutaneous formulation. Evaluate using rheology or dynamic light-scattering.
- **Thermal Stress:** Elevated temperatures (e.g., during viral inactivation) can promote protein denaturation and aggregation. Effects are difficult to quantify and should be controlled.
- **High Ionic Strength:** Use of high ionic strength eluents (e.g., in ion exchange or hydrophobic interaction chromatography) may induce aggregation.
- **pH Sensitivity:** Eluents with pH near or far from the protein's isoelectric point can promote aggregation.
- **Filtration Stress:** Filtration processes may lead to aggregation due to membrane contact, air bubbles, or shear stress. Ultrafiltration at high protein concentrations adds risk, especially before filling or during buffer mixing.

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- **Formulation Handling:** Aggregation can be triggered by agitation, freeze-thaw cycles, and prolonged storage of bulk solutions before formulation.
- **Air-Water Interfaces:** Exposure to air-water interfaces during formulation can reduce protein stability and initiate aggregation pathways. Visual inspection may be required.

Note: For factors affecting reworking, refer to **Section 5.9**.

Failure to manage these factors may result in out-of-specification (OOS) drug substance or drug product, visible particulates, reduced storage stability, loss of efficacy, and potentially increased immunogenicity risk ***(17, 18)***. Therefore, before performing any reprocessing, evaluating and understanding the potential impact on product quality is essential. If a reactive reprocessing strategy is used, the sampling plan should include specific analytical tests to demonstrate that reprocessing does not adversely affect the quality, safety, or efficacy of the drug substance or drug product.

Note: For reprocessing, product release may occur in parallel with ongoing stability studies, since the process remains within the validated manufacturing design space. For reworking, stability data must be available prior to release to define retesting intervals, as the introduction of new steps is equivalent to a modified synthesis with potentially higher risk (see **Section 5.0**).

3.6 General Steps to Consider for a Proactive Design and Validation of a Reprocessing Step

A general outline for designing, validating, and filing a reprocessing procedure with the regulators using a proactive approach is provided in this section. **Figure 3.6-1** provides an example of a generic sequence of activities that could support a proactive reprocessing approach. **Steps 1-20** provide an illustration of reprocessing a biopharmaceutical product.

Section 6.0 (Appendix I) provides a comprehensive example of an approach to reprocessing using a hypothetical chromatography step that was created for TR 74 based on the publication *A-Mab: A Case Study in Bioprocess Development* ***(19)***.

In the ideal situation, a manufacturer would plan for reprocessing during development prior to licensure of the product. However, it may also be possible to prospectively define a new step using supplementary submissions created for reprocessing in manufacturing.

The following 20 steps are presented as recommended considerations for evaluating and implementing a reprocessing strategy for a biopharmaceutical product. While the steps are written predicated on a product that has already received marketing approval, many of the key concepts are equally applicable during the development of reprocessing step(s) for a new biopharmaceutical product. Importantly, these steps are not intended to prescribe a fixed sequence but may be adapted to align with the company's quality system, regulatory commitments, and the stage of the product lifecycle.

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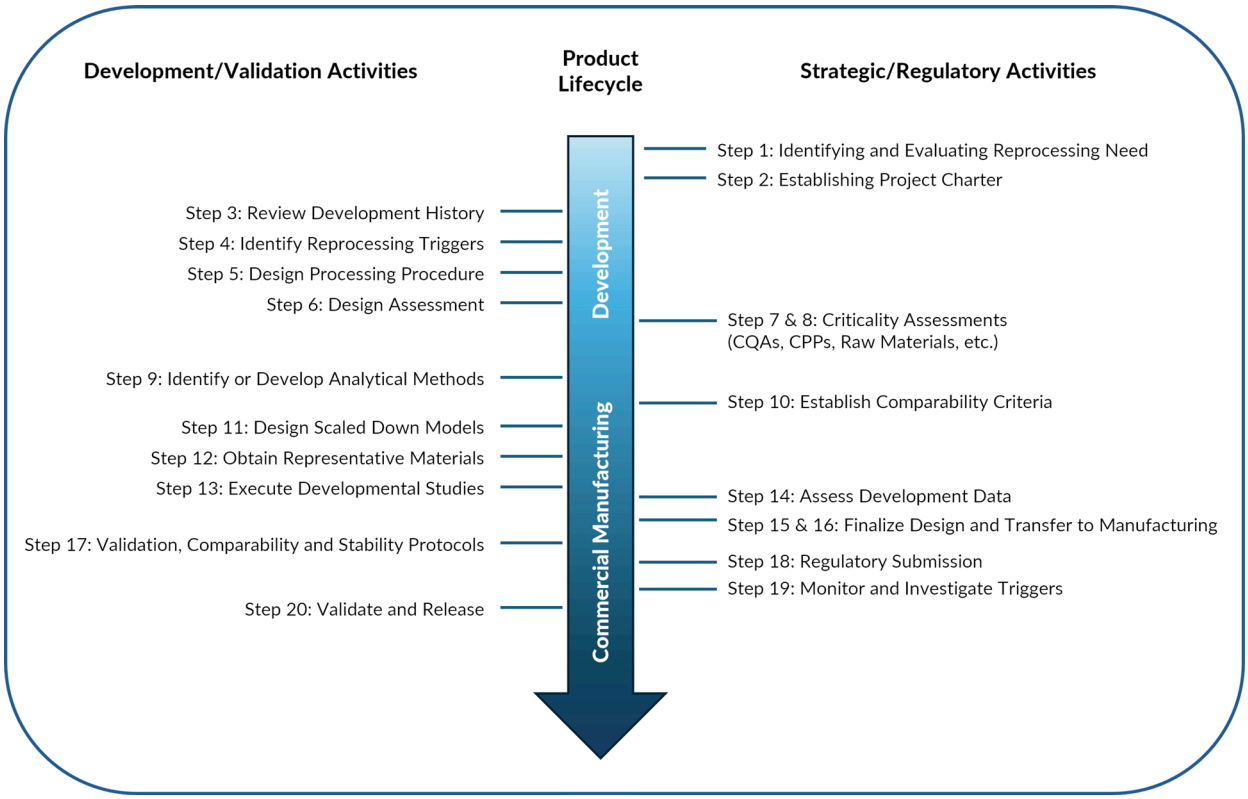

Figure 3.6-1 Sequence of Events – Proactive Reprocessing

Step 1: Identifying and Evaluating Reprocessing Need

The need to reprocess may arise when a biological intermediate/ drug substance/product does not conform to specifications or acceptable ranges. Perform an evaluation to narrow the field of feasible and practical reprocessing steps. Consider that any reprocessing step candidate should be based on simplicity of the step to operate in manufacturing, and past failures. Conduct a technical risk assessment (e.g., FMEA) that includes review of related past manufacturing or development issues that may be helpful in identifying a potential need for reprocessing. Also, consider the effect on drug supply and whether reprocessing is needed to prevent a drug shortage, especially if it is a breakthrough product or an orphan drug. Ensure that further development of a prospective reprocessing step would result in an acceptably low regulatory or quality risk. (An initial quality and regulatory risk assessment is part of the evaluation discussed in **Step 14** and **Step 18**.)

For proactive reprocessing, development data must support that the variability is a rare event and is scientifically understood. (See **Section 3.1**.)

Step 2: Establish a Project Charter (Optional)

Establish a project team and team charter if the evaluation determines the risks to further develop the prospective reprocessing step(s) are acceptable. Development of a reprocessing step involves many stakeholders and may require numerous, varying resources and skillsets including those from the Research and Development, Process/Analytical Development, Manufacturing, Manufacturing Science and Technology (MSAT)/Process Science, Engineering, Quality Control, Quality Assurance, Regulatory Affairs, and Project Management groups.

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Step 3: Review Development History

Review the design basis and development history for the existing unit operation to understand the designed purpose of the proposed reprocessing steps, especially the control of CQAs. Leverage knowledge obtained during the original development of the unit operation to ensure that reprocessing will address the designed purpose of that step to consistently control specific product quality aspects (CQAs) ***(8, 9)***. Perform a systematic assessment to identify which development studies (e.g., viral/impurity clearance, process characterization, resin/membrane cleaning) should be repeated or augmented to support reprocessing. Evaluate the assessment for any potential impact of reprocessing to subsequent processing steps.

Step 4: Identify Reprocessing Triggers

Identify the in-process testing or monitoring triggers that would indicate a need for reprocessing at a given unit operation. It may be necessary to introduce a new in-process test(s) to detect failure modes identified in the initial reprocessing risk assessment. The detection method for a failure mode that leads to reprocessing is critical to the reliability of the overall process and to the success of a reprocessing step. The method must be rapid enough to allow for timely decision-making before the opportunity to reprocess has passed. For example, integrity testing of a 0.2 µm filter would be a better detection method to support reprocessing because it can be completed in a few minutes or hours compared to microbial testing, which can take several days to complete.

Step 5: Design Reprocessing Procedure

Begin the initial design with the proposed manufacturing-scale reprocessing procedure including all equipment, process parameters, linkages to adjacent unit operations, pre-use/post-use procedures (e.g., equilibration/cleaning), and process controls. Give special attention to any procedure required to return the process intermediate and equipment to an appropriate starting condition so that the reprocessing step can be performed. This restoration step is designed to replicate the intermediate input feed-stream conditions (e.g., concentration, volume, pH, conductivity) and is not considered to be a rework step.

Step 6: Design Assessment

Consider the broader effect of the reprocessing design outside the scope of its specific unit operation; for example, facility design, pre-/post-viral segregation, linkages to adjacent unit operations, draw on utilities, and scheduling.

Step 7: Perform Quality Attribute Impact Assessment

Perform a risk assessment ***(6)***, to identify all quality attributes that may be affected by the proposed reprocessing step. In particular, consider such issues as modes of degradation, critical process parameters (CPP), exposure to product-contact surfaces, leachables from filters/single-use tubing, and adsorptive losses that may be impacted by an additional processing step, intermediate hold point, or freeze-thaw cycle. It is not expected that any new CQAs will be identified, but CQAs may take on a higher or lower overall risk following reprocessing ***(8, 9, 20, 21)***. Consider whether the stability of the reprocessed batch should be formally evaluated, especially upon the first instance of reprocessing.

Step 8: Reassess Criticality

Review existing criticality assessments conducted for process parameters, raw materials, equipment, in-process controls, and other factors based on the standard process to determine whether any changes or additions are required. The criticality may be influenced by the addition of a reprocessing step.

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Step 9: Identify or Develop Analytical Methods

Identify or develop the analytical methods to monitor all relevant CQAs in intermediate feed-streams and/or drug substance/product. Even though no new CQAs are likely to be identified to effectively develop and validate a reprocessing step, additional analytical methods capable of monitoring all potentially impacted CQAs in the relevant matrix may be needed to effectively develop and validate a reprocessing step. Additional development and qualification may be needed to adapt methods to the reprocessed product. The risk assessment (**Step 7**) should determine whether additional characterization methods are required to establish comparability and acceptable product quality.

Step 10: Establish Comparability Criteria

Establish appropriate comparability criteria for all CQAs. Once CQAs and their corresponding analytical methods are identified, it may be possible to develop the key comparability criteria to assess product quality during the development of the reprocessing unit operation. All of the original drug substance or drug product specifications must continue to be met following reprocessing. Verify that product quality at intermediate points in the process meets original expectations or consider new criteria as appropriate. Use development studies (discussed below in **Step 11**) when needed to demonstrate that viral or impurity clearance downstream from the reprocessing unit operation is consistent with the original process.

Step 11: Design Scaled-Down Model

Design and qualify a scaled-down model, if applicable. To generate additional data, a scaled-down unit operation model can be adapted from the previously qualified model of the standard process. The requirements for a laboratory-scale model of a reprocessing step are the same as those used in the development of a standard manufacturing process (i.e., design a scaled-down model that is representative of the manufacturing-scale reprocessing unit operation).

Step 12: Obtain Representative Materials

Obtain representative materials for performing reprocessing at the reduced scale. If representative intermediate feed-streams from manufacturing cannot be obtained, then it may be necessary to generate the materials using a qualified scaled-down model of all, or a section of, the manufacturing-scale process. It should also be possible to recreate the failure mode in the feed-stream in a representative manner (e.g., spiking with impurities). If the failure mode is based on operational checks, such as integrity tests, rather than a product-quality excursion, then use of a standard intermediate feed-stream may be justified.

Step 13: Execute Developmental Studies

Plan and execute reduced-scale development studies, if required once the process design elements described above in **Step 4 – Step 12** are completed. The design of reprocessing development studies (e.g., using a design-of-experiments – methodology or worst-case conditions) should result in an understanding of robustness across appropriate critical and noncritical process parameter ranges. Evaluate any effects of the reprocessing step on quality attributes using the selected analytical methods.

The design and scope of these studies should take into consideration all elements of the original development studies related to the unit operation, including intermediate and drug substance/product stability. Determine which prior development studies should be repeated and what additional studies (e.g., testing for new degradants) are needed. The data generated in reduced scale studies will become an important element of regulatory submissions and may reduce the data needed from the manufacturing scale.

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Step 14: Assess Development Data

Upon completion of the development studies, review the generated data to evaluate whether the scaled-down process studies indicate the following:

- All predetermined comparability criteria are met by the reprocessing procedure.
- Process parameter operating ranges are confirmed.
- Process intermediate, drug substance, and product stability are potentially impacted.

Step 15: Finalize Designs

Adapt GMP manufacturing-scale reprocessing procedures and equipment based upon the results of scaled-down studies. For example, if the development studies described in **Step 13** indicate that process parameter ranges must be narrowed or shifted to ensure that all quality and comparability criteria are met, then it may be necessary to modify the manufacturing-scale process design and control strategy to reflect this new process understanding.

Step 16: Transfer Technology to Manufacturing

Execute technology transfer according to company policies and procedures. Transfer of the reprocessing step to manufacturing should consider scale-up issues and implementation of associated GMP procedures. For more complex reprocessing activities, an intermediate pilot-scale process or non-GMP "engineering run" may also be beneficial to ensure a successful transfer into the GMP manufacturing environment (see **Section 3.8** for more information on contamination control strategies).

Step 17: Validation, Comparability and Stability Protocols

Write formal prospective process validation, comparability, and stability protocols for manufacturing-scale reprocessing. Validation considerations include the number of lots of resin or other raw materials required, the number of runs needed, and the additional nonroutine sampling and testing, beginning with consideration of the original validation of the standard process. Additional validation and comparability run(s) at manufacturing scale may be supported by the reduced-scale data generated during development. It is unlikely that these protocols will be executed before the regulatory submission but, if they are, then validation and comparability results could be included in the regulatory submissions discussed in **Step 18** ***(10, 22)***.

Real-time stability studies are considered to be a holistic test of recombinant proteins and, therefore, are often used to document that no changes to the protein have occurred. After validation of the reprocessed unit operation and subsequent license approval, it may not be necessary for reprocessed lots to undergo further stability testing. Ensure the expected stability studies after license approval are clearly stated in the submission. Discuss stability requirements, which may vary regionally, with the appropriate regulatory body.

Accelerated and/or stress stability studies for drug substances may be valuable tools for more rapidly characterizing changes in a protein that may affect stability. In this case, a comparison of accelerated stability data with non-reprocessed control materials, as well as with historical data, may be considered.

Step 18: Regulatory Submission

For approval of the reprocessing step, include the supportive scaled-down development data and prospective validation/comparability/stability protocols in the submission ***(3, 8, 9)***. (The cited publications will be useful references to support the submission.)

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If using the proactive approach, file these elements in the initial marketing application:

- Output of the quality attribute impact assessment (**Step 7**), the criticality reassessment (**Step 8**), and results of the development studies (**Step 13**) in Module 3.2.S.2.6
- Manufacturing-scale reprocessing protocols to execute as needed in Module 3.2.S.2.5, including:
  - Triggers (process failure modes) for reprocessing
  - Relevant tests and acceptance criteria to demonstrate product comparability
  - Stability commitments/protocols
- Any other agency commitments or strategies for implementation (e.g., agency notice of validation completion)
- Description of the validated reprocessing step in Module 3.2.S.2.2

Note: For more details on regulatory guidance, refer to **Section 4.0**.

Step 19: Monitor and Investigate Triggering Event

Monitor failure modes for signs that the manufacturing process might need reprocessing or, alternatively, decide to proactively carry out reprocessing to collect data for a submission. Regardless of whether the reprocessing is prospectively designed, it is essential to thoroughly investigate the root cause of any triggering event and its impact on product quality following GMP procedures and the principles of International Council for Harmonisation (ICH) *Quality Guideline Q9(R1): Quality Risk Management* ***(6)***. The investigation outcome, along with CAPAs, should be considered in relation to process validation, regulatory commitments (**Step 20**), and reprocessing design (**Step 5** and **Step 6**) ***(23)***.

Step 20: Validate and Release

For each reprocessing batch, follow established GMP procedures for executing validation, comparability, and stability protocols, and ensuring all acceptance criteria, including release-testing, are met. Document validation and comparability results in a final report, completing change control and regulatory commitments before releasing the reprocessed batch.

An example for UF/DF reprocessing is in **Table 3.6-1**. During the UF/DF operation, the batch did not reach the target conductivity after the initial diafiltration due to a sustained low transmembrane pressure.

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Table 3.6-1 Example of Reprocessing for Ultrafiltration/Diafiltration

Section	Details
Example Scenario	UF/DF concentration endpoint not met or DF buffer exchange incomplete.
Context	Bulk drug substance undergoes UF/DF to reach target concentration and buffer composition. In process testing shows protein concentration below target and/or conductivity/pH outside specified range after planned DF volume.
Trigger for Reprocessing	In process control indicates endpoint not achieved (e.g., low concentration, buffer exchange incomplete). No evidence of product damage, contamination, or membrane integrity failure. Deviation recorded; immediate assessment indicates issue is correctable without impacting CQAs.
Pre-Reprocessing Assessment	Document deviation, batch status, and hold conditions. Perform risk assessment focusing on CQAs (purity, aggregates, potency, HCP/DNA, residuals). Verify equipment status (e.g., UF module integrity test) and confirm reprocessing is covered by a pre-approved procedure or requires QA authorization.
Reprocessing Plan Objective	Complete buffer exchange and/or achieve final concentration while staying within validated ranges and maximum process exposure (e.g., total shear/recirculation time).
Reprocessing Plan Actions	Resume controlled recirculation to reach target concentration. Perform additional DF cycles with qualified buffer to meet conductivity/pH endpoints. If flux decline/fouling is suspected, consider membrane change under validated procedure (with integrity test and line clearance).
Reprocessing Plan Stop Criteria	Maximum permitted additional DF volume or cumulative processing time; acceptable ranges for concentration, conductivity, pH; stability/exposure limits (e.g., temperature, shear).

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Section	Details
Controls and Documentation	Record all actions, times, volumes, in-process results, equipment IDs. Maintain traceability of buffers and materials used during reprocessing. Ensure reprocessing stays within validated process boundaries; if not, escalate to QA/QC and follow rework or rejection pathways.
Post-Reprocessing Verification	Perform required in process and release testing to confirm CQAs/specifications (e.g., concentration, purity profile, aggregates, bioburden/endotoxin, residuals). Document outcomes, deviations, investigations, and CAPA as applicable.
Release Decision	If acceptance criteria are met and reprocessing adhered to approved procedure/validation limits, proceed to next step or release. If criteria are not met, follow defined pathways for additional investigation, potential rework (if allowed), or batch rejection.
Regulatory and Quality Notes	Reprocessing in UF/DF is permissible if pre-defined, validated, and does not adversely affect product quality. Expect QA oversight and documentation of rationale, risk assessment, and verification data. Establish validation limits (e.g., max additional DF cycles, cumulative processing time, membrane changes).

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3.7 Regulatory Submission for Reactive Approach

Similar to the proactive approach, consider the following elements in the supplemental marketing application for a reactive reprocess:

1. Output of the quality attribute impact assessment (**Step 7**), the criticality reassessment (**Step 8**), and the results of the development studies (**Step 13**) in Module 3.2.S.2.6
2. Manufacturing-scale reprocessing execution results, including:
   - The trigger (process failure mode) resulting in reprocessing
   - Results of any additional full-scale testing with comparison to acceptance criteria
   - Stability protocols and any interim stability data
3. Updated Module 3.2.S.2.2, including a description of the validated reprocessing step

In parallel to filing a supplemental filing for the process change to include an additional reprocessing step, the manufacturer needs to work directly with FDA to obtain approval for batch use.

3.8 Impact of Contamination Control Strategy during Reprocessing

As discussed above, it is critical to ensure microbial controls are considered and protected during reprocessing events. The reprocessed biological drug substances and sterile drug products should adhere to all CQAs associated with bioburden and endotoxins.

During the reprocessing of a sterile biological drug substance and drug product, European Union (EU) Annex 1 ***(20)***, provides general guidance that should be used in the design and control of facilities, equipment, systems, and procedures used for the manufacture of all sterile products. The principles of quality risk management (QRM) should be applied to ensure that microbial, particulate, and endotoxin/pyrogen contamination is prevented in the final drug substance and drug products. QRM is also applicable to low-bioburden biological intermediates and drug substances ***(20)***.

Critical aspects considered in QRM include, but are not limited to:

- Process step where reprocessing occurs
- Time of product exposure (open operations)
- Air classification of the reprocessing work zone (where the product is exposed in absence of closed system)
- Particulate counts and bioburden trends
- Training records of personnel carrying out reprocessing
- Sanitization and disinfection procedures
- Material movement and area segregation controls (e.g. CCS impact, pre/post viral areas)
- Hold times during reprocessing
- Line clearance before and after reprocessing
- Cleaning and/or change over limits, in cases of multiproduct facilities
- Filter validation studies (critical on operating ranges while handling concentrated protein solutions)
- Sampling techniques and number of samples taken for studies

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The Quality Control Unit should establish a more comprehensive monitoring program for operators involved in reprocessing operations, which are especially labor-intensive and require complex manipulations and time criticality.

Processes, equipment, facilities, and manufacturing activities should be managed in accordance with QRM principles to provide a proactive means of identifying, scientifically evaluating, and controlling potential risks to quality. Where alternative approaches are used, these should be supported by appropriate rationale, risk assessment, and mitigation and should adhere principles of CCS. A combination of QRM and a contamination control strategy (CCS) should be established for a robust assurance of contamination prevention during reprocessing. The CCS should be actively reviewed and, where appropriate, updated, and it should drive continual improvement of the manufacturing and control methods. (See PDA *Technical Report No. 90: Contamination Control Strategy Development in Pharmaceutical Manufacturing* for further details ***(21, 24)***, *Contamination Control Strategies (CCS) in Low Bioburden Biologic Drug Substance Manufacturing* ***(25)***, and *Contamination Control Strategy Guidance* ***(26)***.)

Common microbiological attributes during reprocessing for biological drug substances and drug products are given in **Table 3.8-1**.

Table 3.8-1 Common Microbiological Attributes during Reprocessing for Biological Drug Substances and Drug Products

Substance/Product	Bioburden	Endotoxin	Sterility	Container Closure Integrity
Low Bioburden (Nonsterile Drug Substance)	1-3 CFU/100 ml (Depends on CQA)	≤0.25 EU/ml	N/A	Yes ***(14)***
Sterile Drug Substance	N/A	≤0.25 EU/ml	Yes	Yes*
Sterile Drug Product	N/A	≤0.25 EU/ml	Yes	Yes*

*CFR Part 211—Current Good Manufacturing Practice for Finished Pharmaceuticals, Section §211.94(b) in Subpart E, and Part 600—Biological Products: General ***(21, 24)***

Note: The above mentioned limits are suggested and each firm should assess their own product specific limits with respect to product quality and patient safety. Each firm must establish product-specific limits based on risk assessment, product characteristics, and patient safety considerations. Justification for alternative limits should be documented and included in regulatory submissions.

With respect to bioburden, endotoxin, sterility, particulates, and container closure integrity, all the reprocessing procedures should be evaluated, reviewed, and approved by the Quality Control Unit, and the impact on validation and stability should be thoroughly evaluated.

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4.0 Regulatory Considerations

Both reprocessing and reworking decisions require regulatory agency approval prior to release of the affected batch(es) for use in humans. **Section 4.0** provides regulatory agency guidance about what is necessary to report and how to submit this information for approval. In addition, experiences and practices from a PDA Technical Report Team representing a cross-section of industry professionals are provided in the effective management of this communication.

For detailed information on reworking, see **Section 5.0**.

4.1 General

Some regulatory agencies provide clear guidance and details on submissions for these scenarios, while others provide little to no direct guidance. In the latter case, the applicant should determine if reprocessing or reworking can be executed under GMP control only. If in doubt, the wisest practice is to seek relevant regulatory agency approval before considering reworking or reprocessing. For products manufactured across global supply networks, harmonization of reprocessing/reworking strategies across sites is critical. Regulatory submission timelines should be aligned with supply chain planning to avoid stock-outs and ensure uninterrupted patient supply.

Regulatory considerations for several regions are presented in **Sections 4.2–4.5.** When intermediates are manufactured by CDMOs, roles and responsibilities must be clearly defined. This includes identifying the stage of processing, the nature of the biological intermediate, and evaluating the potential impact of reprocessing or reworking. CDMOs should ensure transparent communication, data integrity, and timely provision of information required for regulatory submissions.

4.2 Guidance for International Council for Harmonisation Regions

ICH has brought together regulatory authorities and representatives from the pharmaceutical industry to discuss scientific and technical aspects of pharmaceuticals. ICH includes regulatory authority members from more than 15 countries (e.g., United States, European Union, Japan, Brazil, China, etc.). At *Step 5* of the ICH process, harmonized ICH guidelines are implemented by ICH Regulatory Members and Observers within their respective country or region. ICH Guidelines are implemented in accordance with the applicable national/local/regional rules. The stage of implementation of all ICH Guidelines is dependent upon when a Member or Observer has joined ICH. Progress in ICH Guideline implementation can be tracked at www.ICH.org ***(27)***.

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Guidelines published by ICH contain harmonized guidance specifically for to-be-marketed and marketed pharmaceuticals, including biologicals, specifically for submissions involving reprocessing of biologicals. Some of the common characteristics harmonized to enable efficient reprocessing submissions include:

- To-Be-Marketed: Initial market application to contain details of any anticipated future reprocessing in the relevant sections of the Common Technical Document (CTD) referred to as a Post-Approval Change Management Protocol (PACMP). Allowing this inclusion of reprocessing protocols for relevant steps provides reprocessing flexibility and minimizes post-approval reporting.
- Marketed: Post-market clear guidance and well-defined timeframes for review and approval of Post-Approval Changes (PAC), as well as PACMPs
- Acceptance of science- and risk-based approaches for developing data to support PACs and PACMPs
- Relatively fewer requests for additional documentation not included in CTD Module 3

Table 4.2-1 provides guidance on reprocessing/reworking in the ICH guidelines.

Table 4.2-1 Regulatory Guidance on Reprocess/Reworking

ICH Guideline	Guidance on Reprocessing/Reworking
ICH Q7: Good Manufacturing Practice Guide for Active Pharmaceutical Ingredients (3)	**Reprocessing** Introducing an intermediate or API, including one that does not conform to standards or specifications, back into the process and reprocessing by repeating a crystallization step or other appropriate chemical or physical manipulation steps (e.g., distillation, filtration, chromatography, milling) that are part of the established manufacturing process is generally considered acceptable. However, if such reprocessing is used for a majority of batches, such reprocessing should be included as part of the standard manufacturing process. Continuation of a process step after an in-process control test has shown that the step is incomplete is considered to be part of the normal process. This is not considered to be reprocessing. Introducing unreacted material back into a process and repeating a chemical reaction is considered to be reprocessing unless it is part of the established process. Such reprocessing should be preceded by careful evaluation to ensure that the quality of the intermediate or API is not adversely impacted due to the potential formation of by-products and over-reacted materials.

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ICH Guideline	Guidance on Reprocessing/Reworking
ICH Q7: Good Manufacturing Practice Guide for Active Pharmaceutical Ingredients (3)	**Reworking** Before a decision is taken to rework batches that do not conform to established standards or specifications, an investigation into the reason for non-conformance should be performed. Batches that have been reworked should be subjected to appropriate evaluation, testing, stability testing if warranted, and documentation to show that the reworked product is of equivalent quality to that produced by the original process. Concurrent validation is often the appropriate validation approach for rework procedures. This allows a protocol to define the rework procedure, how it will be carried out, and the expected results. If there is only one batch to be reworked, then a report can be written and the batch released once it is found to be acceptable. Procedures should provide for comparing the impurity profile of each reworked batch against batches manufactured by the established process. Where routine analytical methods are inadequate to characterize the reworked batch, additional methods should be used.
ICH Q7: Questions and Answers (9)	Does the definition of expiry date in ICH Q7 preclude the rework or reprocess of an expired API? According to the definition, material should not be used after the expiry date. The original intent of this definition in ICH Q7 was that expired API should not be used in drug product formulation. It may be acceptable to reprocess or rework the expired API where the API manufacturer has all related historical GMP documentation and additional stability data on the reworked or reprocessed API. There may be registration/filing considerations that are beyond the scope of ICH Q7 in addition to the GMP considerations.
ICH Q9(R1): Quality Risk Management (6)	Quality Risk Management as Part of Materials Management Assessment and evaluation of suppliers and contract manufacturers. Use of materials. To determine appropriateness of reprocessing, reworking, use of returned goods.
ICH Q13: Continuous Manufacturing of Drug Substances and Drug Products (28)	Equipment Design and System Integration Filtration steps in CM may be subject to longer filtration periods and potentially increased throughput per unit area or a greater number of filter changes than those under batch manufacturing. Given these factors, a control strategy and a clearly defined scheme should be put in place to allow for filter changes and pre- and post-use integrity testing, as appropriate, without interrupting the process. In the event of a filter failure, a clear strategy for material diversion and refiltration (reprocessing) should be defined.

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4.3 Specific Guidance from the United States (FDA)

In addition to implementation of the ICH Guidelines, the FDA provides general guidance for biologicals during the clinical investigation period and specific guidance for biologicals during the post-market approval period.

4.3.1 During Clinical Investigation Period

In 2008, FDA published *Guidance for Industry: CGMP for Phase 1 Investigational Drugs* that stressed the importance of documenting how the manufacturing process was carried out during the clinical investigational period ***(29)***:

- A record of manufacturing data that details the materials, equipment, procedures used, and any problems encountered during manufacturing. We recommend that manufacturers retain records sufficient to replicate the manufacturing process. Similarly, if the manufacture of a phase 1 investigational drug batch is initiated but not completed, we recommend that the record include an explanation of why manufacturing was terminated.
- A record of changes in procedures and processes used for subsequent batches along with the rationale for any changes.

Since reprocessing and/or reworking are significant changes, they are required to be reported to the FDA under an information amendment to the Investigational New Drug (IND) Application on file to carry out the human clinical trials (21 CFR 312.31). Although there is no required time period of delay of implementation for these information amendments, manufacturers typically wait for a period of time to allow the FDA reviewers opportunity to review the amendment ***(30)***.

Note: Consider the impact of changes to be discussed with regulatory authorities in order to determine the need for additional toxicology studies.

FDA in 2020 published Guidance for Industry: Chemistry, Manufacturing, and Control (CMC) Information for Human Gene Therapy Investigational New Drug Applications (INDs), the scope of which included all gene therapies, including ex-vivo genetically modified human cells. There is no mention of reworking in this guidance; however, it provides the following guidance for reprocessing ***(31)***:

- **Drug Substance:** For early stage INDs, there may be differences between the manufacturing and testing of the toxicology lots and the material you plan to use in the clinical studies. For later stage INDs, there may be changes to the manufacturing process as part of process development, optimization, or under certain conditions there may be reprocessing step(s). In both situations, we recommend that you describe how manufacturing differences are expected to impact product safety and activity and to provide batch analysis information in the "Batch Analysis (3.2.S.4.4)" section of the CTD.
- **Drug Product:** Under certain conditions, repeating sterile filtration (if applicable) may be critical. We recommend that you clearly describe the conditions and justifications for each reprocessing procedure ***(32)*** and demonstrate product consistency between reprocessed lots and normal production lots.

4.3.2 Post-Market Approval Period

Under 21 CFR 601.12, each post-approval change in the production process established in the approved BLA must be reported using the submission type associated with one of the three tier-based reporting

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categories. This depends on the potential (i.e., risk) of the change to have an adverse effect on the identity, strength, quality, purity, or potency of the product as they may relate to the safety or effectiveness of the product (i.e., a substantial potential, moderate potential, or minimal potential). The reporting categories in this regulatory provision also specify when the product made using the change can be distributed. The submission types and associated reporting categories are as follows ***(33)***:

- **Prior Approval Supplement (PAS)**
  Change that has a substantial potential to have an adverse effect on product quality (i.e., major changes) require an applicant to report the change to the FDA in a supplement to the approved BLA. A PAS must be approved by the FDA prior to distribution of the product manufactured using the change (21 CFR 601.12(b).
- **Changes Being Effected in 30 Days/Supplement–Changes Being Effected (CBE30/CBE)**
  Changes that have a moderate potential to have an adverse effect on product quality (i.e., moderate changes) require an applicant to report the change to the FDA in a supplement at least 30 days prior to distribution of the product made using the change (21 CFR 601.12(c)).
- **Annual Report (AR)**
  Changes that have a minimal potential to have an adverse effect on product quality (i.e., minor changes) shall be documented by the applicant in an annual report (21 CFR 601.12(d)).

In 1997, FDA published Guidance for Industry: Changes to an Approved Application for Specified Biotechnology and Specified Synthetic Biological Products, the scope of which included recombinant DNA-derived protein/polypeptide products approved under the Federal Food, Drug, and Cosmetic Act (FDCA) and complexes or conjugates of a drug with a monoclonal antibody approved under the FDCA. There is no mention of reworking in this guidance; however, it provides guidance specifically noting that reprocessing without a previously approved reprocessing protocol is considered a change with substantial potential to adversely affect product quality, safety, or effectiveness ***(34)***.

In 2021, FDA published *Guidance for Industry: Chemistry, Manufacturing, and Controls Changes to an Approved Application: Certain Biological Products***,** the scope of which included vaccines, plasma-derived protein products (and their recombinant analogs), cellular and gene therapy products, among others. **Table 4.3.2-1** lists the guidance provided for both reprocessing and reworking ***(12)***.

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Table 4.3.2-1 Guidance for Reprocessing/Reworking in CMC Changes to Approved Application

Prior Approval Supplement (PAS)	CBE30/CBE	Annual Report (AR)
Drug Substance		
Reprocessing in the manufacture of a drug substance without an approved protocol	Yes	Reprocessing in the manufacture of a drug substance without an approved protocol
Reprocessing in the manufacture of a drug substance with an approved protocol	N/A	Reprocessing in the manufacture of a drug substance with an approved protocol
Reworking in the manufacture of a drug substance	N/A	N/A
Drug Product		
Reprocessing in the manufacture of a drug product without an approved protocol	Yes	Reprocessing in the manufacture of a drug product without an approved protocol
Reprocessing in the manufacture of a drug product with an approved protocol	N/A	Reprocessing in the manufacture of a drug product with an approved protocol
Reworking in the manufacture of a drug product	N/A	N/A

4.4 Specific Guidance from the European Union (EMA)

In addition to implementation of the ICH Guidelines, EMA provides specific guidance for biologicals both during the clinical investigation period and the post-market approval period.

4.4.1 During Clinical Investigation Period

EMA in 2022 published the *Guideline on the Requirements for Quality Documentation Concerning Biological Investigational Medicinal Products in Clinical Trials – Revision 2*. Included in the scope were proteins and polypeptides (produced from recombinant and nonrecombinant cell-culture expression systems), their derivatives and products of which they are components (e.g., conjugates), and auxiliary medicinal products containing these proteins and polypeptides as active substances. There is no mention of reworking in this guideline; however, the following guidance is provided for reprocessing of these biological manufacturing processes ***(35)***:

- **Drug Substance:** Any reprocessing during manufacture of the active substance (e.g. filter integrity test failure) should be described and justified. Reprocessing could be considered in exceptional circumstances. For biological products, these situations are usually restricted to certain refiltration and re-concentration steps upon technical failure of equipment or mechanical breakdown of a chromatography column.
- **Drug Product:** Reprocessing may be acceptable for particular manufacturing steps (e.g. refiltration) only if the steps are adequately described and appropriately justified.

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A change to the production process submitted in the investigational medicinal product dossier (IMPD), if it is a substantial modification, must be reported. Substantial modification means any change that is likely to have a substantial impact on the safety and rights of the subjects or on the reliability and robustness of the data generated in the clinical trial. **Table 4.4.1-1** provides guidance for the reporting of reprocessing of biological manufacturing processes.

Table 4.4.1-1 Guidance for Reporting Reprocessing of Biological Processes

Substantial Modification	Non-Substantial Modification
Manufacturing process of the active substance: any reprocessing not described in the IMPD	Manufacturing process of the active substance: reprocessing if adequately described and accepted in the initial submission

In 2025, EMA published the *Guideline on Quality, Non-Clinical and Clinical Requirements for Investigational Advanced Therapy Medicinal Products in Clinical Trials* the scope of which includes gene therapy medicinal products, somatic cell therapy medicinal products, tissue-engineered products, and combined ATMPs. There is no mention of reworking in this guideline; however, the following guidance is provided for reprocessing of these biological manufacturing processes ***(36)***:

- **Drug Substance:** Any reprocessing during manufacture of the active substance (e.g., filter integrity test failure) should be described and justified. Reprocessing could only be considered in exceptional circumstances. These situations are usually restricted to refiltration and re-concentration steps upon technical failure of equipment or mechanical breakdown of a chromatography column.
- **Drug Product:** Reprocessing may be acceptable for particular manufacturing steps (e.g. refiltration) only if the steps are adequately described and appropriately justified.

4.4.2 Post-Market Approval Period

Under the EMA Variation Regulation, each post-approval change in the production process established in the approved Marketing Authorisation Application must be reported using the submission type associated with one of the three tier-based reporting categories. The category assigned depends on the potential (i.e., risk) of the change to have an adverse effect on the identity, strength, quality, purity, or potency of the product as they may relate to the safety or effectiveness of the product (i.e., a substantial potential, moderate potential, or minimal potential). The reporting categories in this regulatory provision also specify when the product made using the change can be distributed. The submission types and associated reporting categories are as follows ***(37)***:

- **Type II Major Variation:** Change must be approved by EMA prior to distribution of the product manufactured using the change
- **Type IB Minor Variation:** Change must wait 30 days after submission to EMA
- **Type IA Minor Variation:** Change must be submitted within 12 months from the date of implementation

There is no mention of reworking in the Variation Regulation. **Table 4.4.2-1** provides guidance for reprocessing biological manufacturing processes.

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Table 4.4.2-1 Guidance on Reprocessing in the EMA Variation Regulation

TYPE II
B.I.a.2 Changes in the manufacturing process of the active substance: c) The change refers to a biological/immunological substance or use of a different chemically derived substance in the manufacture of a biological/immunological substance, which may have a significant impact on the quality, safety and efficacy of the medicinal product and is not related to a protocol
B.II.b.3 Change in the manufacturing process of the finished product, including an intermediate used in the manufacture of the finished product: c) The product is a biological/immunological medicinal product and the change requires an assessment of comparability

4.5 Other Guidance

Biological manufacturers wish to operate across many geographical areas, yet not all countries participate in ICH (e.g., Russia). Keeping track of each country's or region's CMC regulatory compliance review procedures and oversite for reprocessing and reworking of biologicals has proven challenging.

Various specialized government agencies and industry associations, such as those listed here, provide recommendations on effective control and safety when considering reprocessing and/or reworking of biological manufacturing processes:

- World Health Organization (WHO) is a United Nations agency to support health around the world, including the safety of manufactured drugs
- Pharmaceutical Inspection Co-Operation Scheme (PIC/S) is an informal, nonbinding cooperative arrangement between regulatory authorities in the field of GMPs for medicinal products
- BioPhorum is a collaboration of industry leaders and subject matter experts working together to pool knowledge, best practices, and ideas on global biopharmaceutical industry practices
- Parenteral Drug Association (PDA) is the leading global provider of science, technology, and regulatory information detailed in technical reports and points-to-consider documents

Overall, carrying out reprocessing or reworking according to the principles indicated in the ICH guidelines and the FDA and EMA guidances is generally a safe approach. However, if there is any doubt about proceeding with reprocessing or reworking a biological manufacturing process, contact the respective regulatory authority for specific guidance.

Note: If reprocessing or reworking has not been explicitly mentioned in the manufacturer's submitted documentation on the manufacturing process, correctly assume that it has not yet been approved.

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5.0 Introduction to Reworking

It is the primary responsibility of the Manufacturing Authorization Holder (MAH) of a biological drug substance and/or drug product consistently meet all quality, safety, purity, and identity requirements. This authoring team has strict recommendations on how and when such a reworking action should be considered by the MAH. The ultimate approval of a reworked drug substance and/or drug product for human use is solely at the discretion of the approving regulatory agency.

ICH and EU define reworking as "Subjecting an intermediate or API that does not conform to standards or specifications to one or more processing steps that are different from the established manufacturing process to obtain acceptable quality intermediate or API" ***(3, 38)***. During reworking, the nonconforming material is subjected to a different manufacturing process to get a product that meets all the predefined acceptance criteria and specifications for quality, safety, purity, and identity (QSIP).

The scope of reworking for biologicals includes a diverse array of therapeutic proteins, each with unique challenges and considerations for manufacturing and quality control including, among others:

- Interferons
- Cytokines
- Growth factors
- Interleukins
- Coagulation factors
- Blood factors
- Monoclonal antibodies (IgM, IgD, IgG, IgE, IgA)
- Bispecific antibodies (IgG-based bsAbs, fragment-based bsAbs)
- Antibody drug conjugates targeting cancer therapy
- Fusion proteins
- Glycoprotein hormones (e.g., rhCG, rFSH)
- Insulins
- Vaccines

Each of these therapeutic categories requires specific attention to factors like protein stability, aggregation prevention, and purification processes to ensure product purity, quality, efficacy, and safety.

Due to the intricate sourcing, including the nature of their manufacturing processes and stability concerns, cell and gene therapies are excluded from the scope for reworking. The complexity of these treatments often involves different considerations that require specialized processes and controls distinct from those applicable to the protein-based biologicals listed above.

Figure 5.0-1 gives an overview of possible areas of reworking for biological drug substances and drug products, applicable ICH guidances, and relevant PDA technical reports that enable documentation to write appropriate control strategies.

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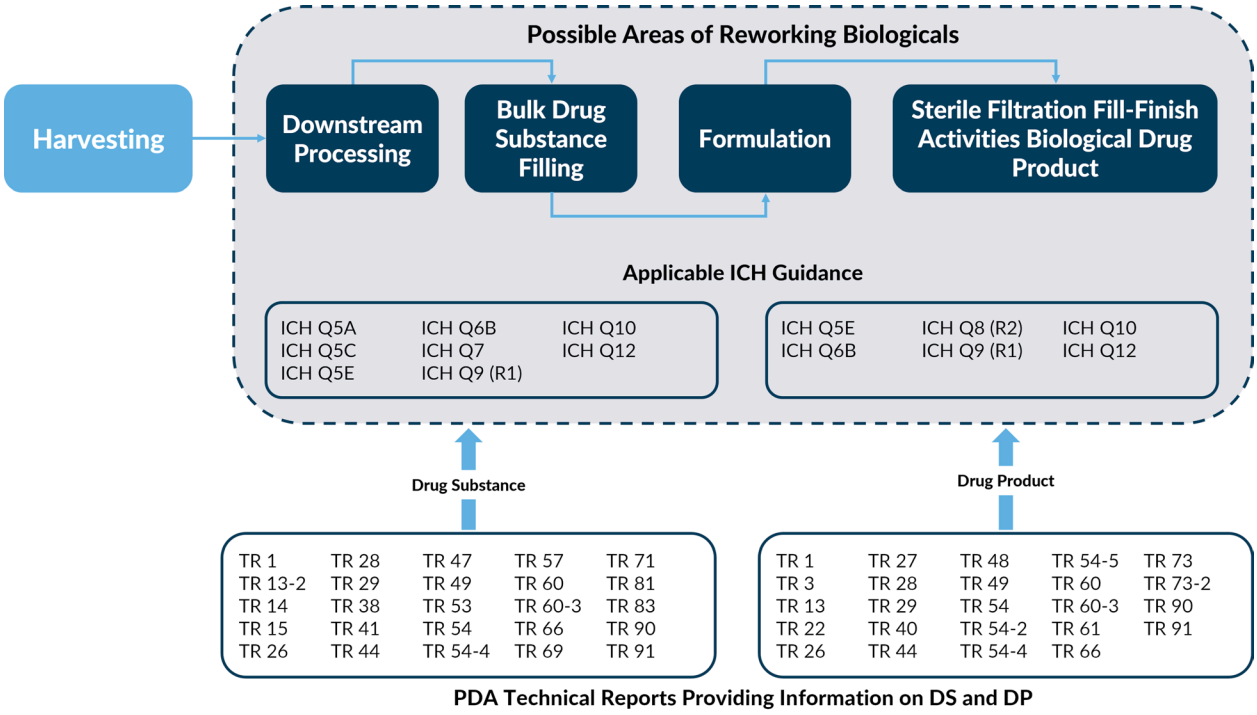

Figure 5.0-1 Overview of Reworking of Biological Drug Substances and Drug Products*

*Titles of the documents referenced in **Figure 5.0-1**:

- *ICH Q5A(R2): Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin*
- *ICH Q5C: Stability Testing of Biotechnological/Biological Products*
- *ICH Q5E: Comparability of Biotechnological/Biological Products Subject to Changes in their Manufacturing Process*
- *ICH Q6B: Specifications: Test Procedures and Acceptance Criteria for Biotechnological/Biological Products*
- *ICH Q7: Good Manufacturing Practice Guide for Active Pharmaceutical Ingredients*
- *ICH Q9(R1): Quality Risk Management*
- *ICH Q10: Pharmaceutical Quality System*
- *ICH Q12: Technical and Regulatory Considerations for Pharmaceutical Product Lifecycle Management*
- *Technical Report No. 1 (Revised 2007): Validation of Moist Heat Sterilization Processes: Cycle Design, Development, Qualification and Ongoing Control*
- *Technical Report No. 3: Validation of Dry Heat Processes Used for Depyrogenation and Sterilization*
- *Technical Report No. 13 (Revised 2022): Fundamentals of an Environmental Monitoring Program*
- *Technical Report No. 13-2: Fundamentals of an Environmental Monitoring Program Annex 1: Environmental Monitoring of Facilities Manufacturing Low Bioburden Products*
- *Technical Report No. 14 (Revised 2008): Validation of Column-Based Chromatography Processes for the Purification of Proteins*
- *Technical Report No. 15 (Revised 2009): Validation of Tangential Flow Filtration in Biopharmaceutical Applications*
- *Technical Report No. 22 (Revised 2025): Process Simulation for Aseptically Filled Products*
- *Technical Report No. 26 (Revised 2025): Sterilizing Filtration of Liquids*
- *Technical Report No. 27: Pharmaceutical Package Integrity*
- *Technical Report No. 28 (Revised 2006): Process Simulation Testing for Sterile Bulk Pharmaceutical Chemicals*
- *Technical Report No. 29 (Revised 2012): Points to Consider for Cleaning Validation*

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- *Technical Report No. 38: Manufacturing Chromatography Systems Postapproval Changes (ChromPAC): Chemistry, Manufacturing, and Controls Documentation*
- *Technical Report No. 40: Sterilization Filtration of Gases*
- *Technical Report No. 41: Virus Filtration*
- *Technical Report No. 44: Quality Risk Management for Aseptic Processes*
- *Technical Report No. 47: Preparation of Virus Spikes Used for Virus Clearance Studies*
- *Technical Report No. 48: Moist Heat Sterilizer Systems: Design, Commissioning, Operation, Qualification and Maintenance*
- *Technical Report No. 49: Points to Consider for Biotechnology Cleaning Validation*
- *Technical Report No. 53: Guidance for Industry: Stability Testing to Support Distribution of New Drug Products*
- *Technical Report No. 54: Implementation of Quality Risk Management For Pharmaceutical and Biotechnology Manufacturing Operations*
- *Technical Report No. 54-2: Implementation of Quality Risk Management for Pharmaceutical and Biotechnology Manufacturing Operations, Annex 1: Case Study Examples for Quality Risk Management in Packaging and Labeling*
- *Technical Report No. 54-4: Implementation of Quality Risk Management for Pharmaceutical and Biotechnology Manufacturing Operations, Annex 3: Case Studies in the Manufacturing of Biotechnological Bulk Drug Substances*
- *Technical Report No. 54-5: Quality Risk Management for the Design, Qualification, and Operation of Manufacturing Systems*
- *Technical Report No. 57: Analytical Method Validation and Transfer for Biotechnology Products*
- *Technical Report No. 60 (Revised 2026): Process Validation: A Lifecycle Approach*
- *Technical Report No. 60-3: Process Validation: A Lifecycle Approach Annex 2: Biopharmaceutical Drug Substances Manufacturing*
- *Technical Report No. 61: Steam in Place*
- *Technical Report No. 66: Application of Single-Use Systems in Pharmaceutical Manufacturing*
- *Technical Report No. 69: Bioburden and Biofilm Management in Pharmaceutical Manufacturing Operations*
- *Technical Report No. 71: Emerging Methods for Virus Detection*
- *Technical Report No. 73: Prefilled Syringe User Requirements for Biotechnology Applications*
- *Technical Report No. 73-2: Application of Medical Device Regulations, Annex I Requirements (GSPRs) for Staked Needle Syringes*
- *Technical Report No. 81: Cell-Based Therapy Control Strategy*
- *Technical Report No. 83: Virus Contamination in Biomanufacturing: Risk Mitigation, Preparedness, and Response*
- *Technical Report No. 90: Contamination Control Strategy Development in Pharmaceutical Manufacturing*
- *Technical Report No. 91: Post-Approval Change Management and Implementation for Biologics and Pharmaceutical Drug Products--A User's Guide*

5.1 Prioritization Tool for Reworking

Using the MoSCoW (Must-have, Should-have, Could-have, and Won't-have, or will not have right now), Prioritization Tool to evaluate reworking of a biological drug substance involves systematically categorizing tasks or issues based on their criticality and impact ***(39)***.

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This step-by-step guide outlines how to apply the MoSCoW method in this context, and **Figure 5.1-1** illustrates the Patient-Centric Reworking Prioritization tool.

1. **Gather Relevant Data and Stakeholders**
   - Collect all information related to the reworking process, including technical assessments, quality data, regulatory considerations, and risk analyses.
   - Assemble a cross-functional team comprising manufacturing, quality assurance, regulatory affairs, and process development experts.
2. **Define the Scope of Reworking**
   - Clearly specify which aspects of the biological drug substance are being considered for rework (e.g., purification, formulation, in-process controls, final product adjustments).
3. **Identify Rework Tasks or Issues**
   - List all potential rework activities, corrections, or modifications needed to meet quality and regulatory standards.
4. **Apply the MoSCoW Prioritization Criteria**
   - Categorize each task or issue into one of the following groups:
     - Must have (M): Critical for human survival, compliance, safety, efficacy, or regulatory approval. These are nonnegotiable and should be completed before proceeding.
     - Should have (S): Immediate medical supply need; can be scheduled after Must-Have items.
     - Could have (C): Desirable to support patient needs, especially those suffering with genetic disorders, cancer, glaucoma, blood issues, and wherever critical to support health condition of work force.
     - Won't have (W): Tasks that are not necessary in the current scope or can be deferred.
5. **Prioritize Reworking Activities**
   - Focus first on Must-Have tasks to ensure the reworked product meets all quality and regulatory standards.
   - Schedule Should-Have activities shortly afterwards, resources permitting.
   - Consider Could-Have tasks, if time and resources allow, to enhance overall process robustness.
   - Leave Won't-Have tasks for future considerations or continuous improvement.
6. **Document and Communicate**
   - Record the prioritization outcomes clearly, including rationales for each category.
   - Communicate the priorities to all relevant teams and stakeholders to ensure aligned execution.
7. **Implement and Monitor**
   - Proceed with rework activities according to the established priorities.
   - Monitor progress and adjust priorities if new information or issues arise.
8. **Additional Measures**
   - Ensure compliance with GMP and regulatory requirements throughout the rework process.
   - Use risk assessments to support the categorization, especially for CQAs.

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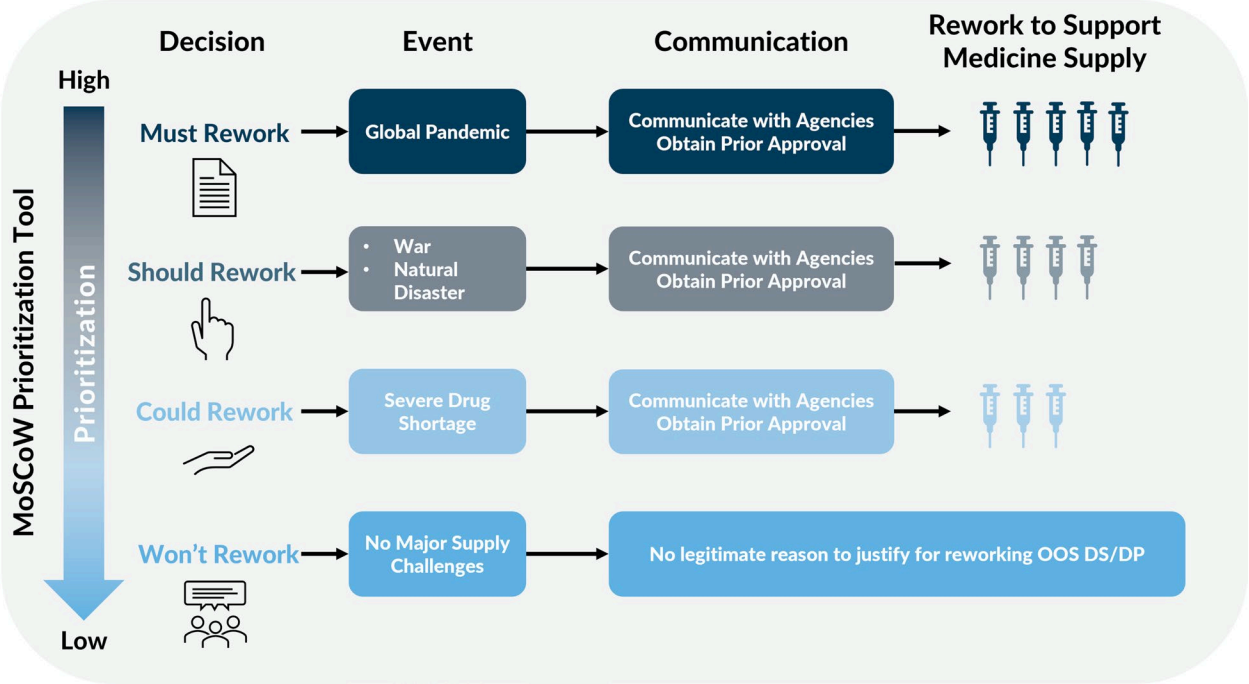

Figure 5.1-1 Must-have, Should-have, Could-have, and Won't-have, or Will Not Have Right Now as a Patient-Centric Prioritization Tool

Systematically applying the MoSCoW prioritization method can effectively focus resources on the most-critical rework activities, ensuring the drug substance and/or drug product meets quality, safety, and regulatory standards efficiently.

Importantly, the prioritization tool mentioned in this technical report serves as an example. The manufacturing organization must align with the company's quality system and utilize approved prioritization tools.

5.2 Difference Between Reprocessing and Reworking

Table 5.2-1 and **Table 5.2-2** outline the major differences between reprocessing (see **Section 3.0**) and reworking (see **Section 5.0**).

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Table 5.2-1 Reprocessing versus Reworking *(40)*

Reprocessing	Reworking
Reprocessing is an exact repetition of one or more steps in the established manufacturing process	Reworking is based on the inclusion of new steps in the established manufacturing process
To support annual product quality review, GMP requirements need to be followed within the QMS to support regulatory approval	To support annual product quality review, GMP requirements need to be followed within the QMS to support regulatory approval
May be approved as part of the manufacturing license, or regulatory approval may be a post-approval supplement (see **Section 3.0**)	Regulatory submission and formal approval are mandatory
Nonconforming batches (including mechanical equipment failures)	Only nonconforming batches (does not include mechanical equipment failures)
Applicable to low-bioburden drug substance and sterile drug substance and/or drug product* ***(23, 41, 42)***	Applicable to low-bioburden drug substance and sterile drug substance and/or drug product* ***(23, 43-46)***
Batch subjected to one or more steps that are part of established manufacturing process	Batch subjected to one or more steps that are different from established manufacturing process
Contamination control strategy must be considered* ***(23, 41, 44, 47)***	Contamination control strategy must be considered* ***(23, 41, 44-47)***
If reprocessing is used for a majority of batches, it should be included as part of the standard manufacturing process	Each reworking event is unique and should not be part of the standard manufacturing process
Reprocessed batches should be subjected to appropriate evaluation	Reworked batches should be subjected to appropriate evaluation, testing, and stability testing, if warranted, with documentation to demonstrate that the reworked batches are of equivalent quality to that produced by the original process* ***(2, 48)***
Concurrent validation is often appropriate	Concurrent validation is often appropriate
*For more information, refer to the references cited.	

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Table 5.2-2 Reprocessing and Reworking for Drug Substance and Drug Product *(40)*

Biological Intermediate, Drug Substance (API)	Drug (Medicinal) Product
Clear distinction between reprocessing and reworking	Often, there is no clear distinction between reprocessing and reworking **Note:** Drug products cannot be reworked, they must be reprocessed (e.g., sterile filtration in case of mechanical failures)
Reprocessing is typical	Reprocessing is atypical
Reprocessing generally improves API quality **Note:** In certain cases, reprocessing may restore or maintain API quality; however, it should not be considered a routine practice and must be scientifically justified on a case-by-case basis.	Reprocessing rarely or possibly does not improve drug product quality except sterility attributes **Note:** A detailed risk assessment is needed at this stage.

5.3 Background to Manufacturing Changes

Reworking is an unexpected occurrence and is not preapproved as part of the marketing authorization. The MAH's Quality Unit should oversee investigations into a nonconforming (OOS) in-process drug, a bulk process intermediate (final biological bulk intermediate), a drug substance of a single batch or lot to an alternate manufacturing process during reworking decisions, prior to release to ensure that the details of rework in question has been submitted to the regulatory agency and formal approval has been obtained. The MAH's Quality Unit should oversee investigations into nonconforming (OOS) bulk process intermediate/ drug substances from a single batch or lot. This oversight is crucial during reworking decisions prior to release. The Quality Unit must ensure that the details of the rework have been reviewed and submitted to the regulatory agency and formal approval has been obtained.

Changes that have a substantial potential to have an adverse effect on product quality (i.e., major variations) require the manufacturer (applicant) to report the change to the FDA in a supplement to the approved BLA. A PAS must be approved by the FDA prior to distribution of the product manufactured using the change to protect the public from products whose identity, strength, quality, purity, potency, safety, or effectiveness may be compromised ***(10, 33, 46)***.

If biological drug products are reworked, they must be reported in a supplement to FDA prior to distribution, unless the reworking was done according to a procedure previously approved by the Agency. The type of notification is based on the potential risk of the change having an adverse effect on the identity, strength, quality, purity, or potency of the product as it may relate to the safety or effectiveness of the product ***(24)***.

The Variation Regulation is supplemented by the European Commission's Guidelines on the Details of the Various Categories of Variations, on the Operation of the Procedures Laid Down in Chapters II, IIa, III and IV of Commission Regulation No 1234/2008 of 24 November 2008. The EU guidelines define a major variation of Type II as one that is not an extension of the Marketing Authorization and that may have a significant impact on the quality, safety, or efficacy of a medicinal product ***(37)***. **Figure 5.3-1** shows

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scientific and risk-based categorization changes and **Table 5.3-1** provides major changes in manufacturing that can be designated as reworking.

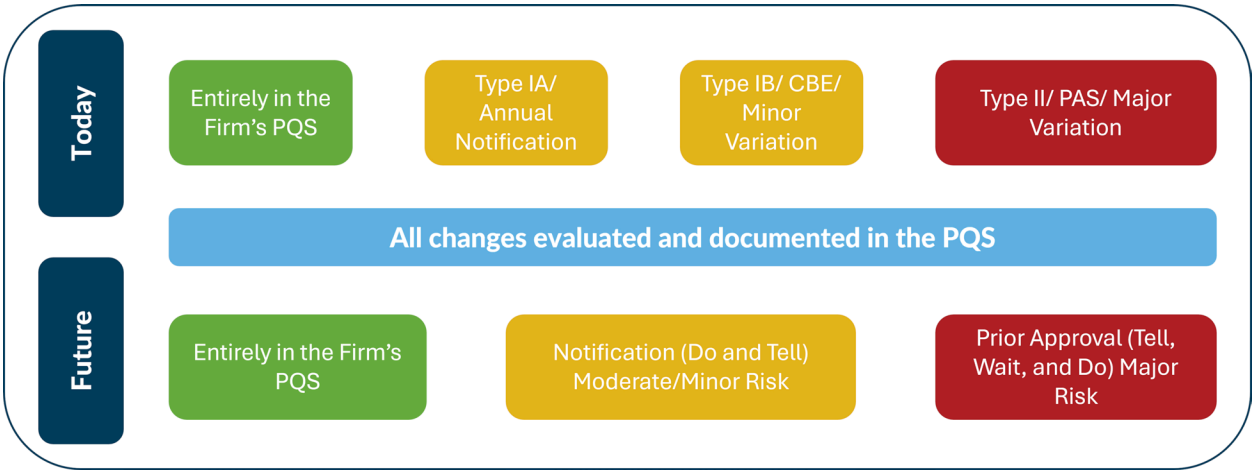

Figure 5.3-1 Scientific and Risk-Based Approach to Categorization of Changes *(49)*

Table 5.3-1 Changes in Manufacturing *(34, 50)*

Position	Major Changes	Approach
Downstream Processing	Changes in viral filtration step Addition/changes/replacement of purification and/or isolation operations Change in concentration of buffers/elution buffers/equilibration buffers and/or changes made during gradient elution steps Change in the order of purification/Isolation and/or polishing steps Changes in hold steps in between processing stages Changes in capacity of purification steps Changes made in Batch/Lot and/or eluate pooling Changes in Chromatography systems ***(51)***	Reworking should be science- and risk-based, considering such elements as the control strategy, comparability, ongoing process validation, and alignment with regulatory approval procedures prior to release of the drug substance or drug product.
Drug Product Manufacturing	Changes in formulation (concentration/excipients/acceptance criteria) Changes in filtration Changes in container closure systems (filling/primary packing materials) Modification of a drug delivery device Changes in shelf life and storage conditions	

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5.4 Proactive Reworking Agreement

Agreeing to changes (or accepted changes made) with regulatory authorities prior to implementation, the MAH may follow the steps for the proactive reworking agreement as discussed in **Table 5.4-1** and **Figure 5.4-1**.

Table 5.4-1 Steps for the Proactive Reworking Agreement

Steps	Information
Step 1	Detection of any in-process nonconformance of a biological intermediate or nonconformance of a drug substance and/or drug product (refer Table 5.3-1 for changes associated with drug product) should be immediately brought to the attention of the Quality Unit and should be escalated to the appropriate level of management (i.e. site quality head). **Note:** The failure of analytical methods or failure to meet any acceptance criteria is considered as nonconformance.
	A detailed investigation or root cause analysis must be performed to understand in depth the exact reason of the nonconformance or failure to meet the acceptance criteria. Before this investigation starts, the analytical tests must be duplicated to confirm the nonconformance.
Step 2	Based on the stage of manufacturing, details, and the extent of nonconformance, after a detailed evaluation, the Quality Unit, cross-functional experts, and management may consider rejecting the nonconforming material/batch/lot or propose a "reworking strategy."
Step 3	The Quality Unit will recommend carrying out a detailed assessment before granting permission to rework the drug substance or drug product that failed to meet specification based on the following preconditions detailed in **Step 4**. The assessment should be based on the application of science, knowledge, and risk approach to understand, evaluate, and manage a successful reworking strategy. Furthermore, the Quality Unit proposes a list of stakeholders and participants required to carry out a unique (one-time) assessment, including a detailed scope, and timeline following GMP format to document observations, concerns, and conclusions against or in favor of the reworking strategy. **Note:** The structural stability of the biological molecule can be compromised if exposed to a longer period in non-validated hold time or conditions. Hence, often the reworking strategy of a drug substance that fails to meet the acceptance criteria can be "rejected" by the Quality Unit. This is due to the impact of the unusually long hold-period conditions of the biological molecule in a nonconformed state during the preliminary investigation steps (between **Step 1** and **Step 3**).
	A detailed review will be performed by the Quality Unit on conclusions of the science, knowledge, and risk assessment and can consider approving a "reworking procedure." The details from the assessment will be captured in the change control as stimulus for reworking.

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Steps	Information
Step 4	An adequate science/risk/knowledge-based strategy must be considered during the drafting of a detailed reworking protocol and its part of a change control activity. All the associated operating procedures, sampling plans, test procedures, and controls should be approved before entering into the draft protocol. All considerations on stages, in-process control tests, enhanced sampling, acceptance criteria, batch/lot numbering, and proposal to store the reworked material under predefined conditions with a detailed strategy on validation (including strategy on small-scale studies) must be included as a part of the "reworking protocol" (PACMP). All the process changes, in-process controls, and additional testing methods should be thoroughly documented in the reworking protocol. The entire reworking protocol and test methods should be reviewed, commented on, and approved by the Quality Unit.
Step 5	Prior Submission Process: The Quality Unit formally approves a change as a part of the reworking protocol, submits the relevant document containing a detailed description on the reason for the change along with justifications, risk assessments, proposed studies, including acceptance criteria, supporting data to be generated during execution of reworking, validation strategy along with a detailed implementation plan within PACMPs on the reported category to the regulatory authority as PAS (FDA) and/or Type II Variation (EMA) ***(52)***. This set of detailed documents must be reviewed and approved by the regulators prior to the execution of the "reworking strategy" for the drug substance and/or drug product that failed to meet the acceptance criteria (OOS) ***(52)***.
Step 6	The regulatory agency reviews the "reworking protocol" (PACMP) plan of change and, including the studies proposed, can request additional details and can ask relevant questions on the change from the MAH during this phase of the Proactive Reworking Agreement (PRA). The MAH provides all the necessary details and answers and agrees to the execution of the preapproved reworking protocol. **Note:** An applicant may ask FDA to expedite its review of a supplement for public health reasons or if a delay in making the change described in it would impose an extraordinary hardship on the applicant. Such a supplement should be plainly marked: "Prior Approval Supplement-Expedited Review Requested" ***(53)***.
Step 7	The reworking protocol should be executed under strict supervision of manufacturing experts and Quality Assurance. The in-process and control samples must be evaluated by Quality Control on a timely basis. The use of PAT tools and documenting the real-time data gives a better overview about the performance of the biological molecule under the reworking conditions ***(54)***. Quality Control should additionally test to ascertain the quality and purity of the reworked material. Additional tests and methods to characterize and compare the reworked materials should be clearly specified.

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Steps	Information
Step 8	A detailed evaluation must be carried on reworked batches to ensure that product quality has not been compromised including, if necessary, using additional analytical test methods and stability testing ***(55, 56)***. Reworking materials and process documents should be reviewed and approved by the Quality Unit, and the impact on validation and stability should be evaluated. Upon a detailed evaluation, the Quality Unit will take a position to proceed with a regulatory submission. The Regulatory Department will make necessary changes and additions in the regulatory documentation and, upon review, will inform to the regulatory authority and submit it as a PAS (FDA) and/or Type II Variation (EMA) ***(52)***
Step 9	After submission of the detailed information and data to the regulatory agency in the prescribed format of reworking protocol, the agency will consider the specific time to review, comment, and ask for further data, if necessary, or approve the reworking for the proposed drug substance and/or drug product.

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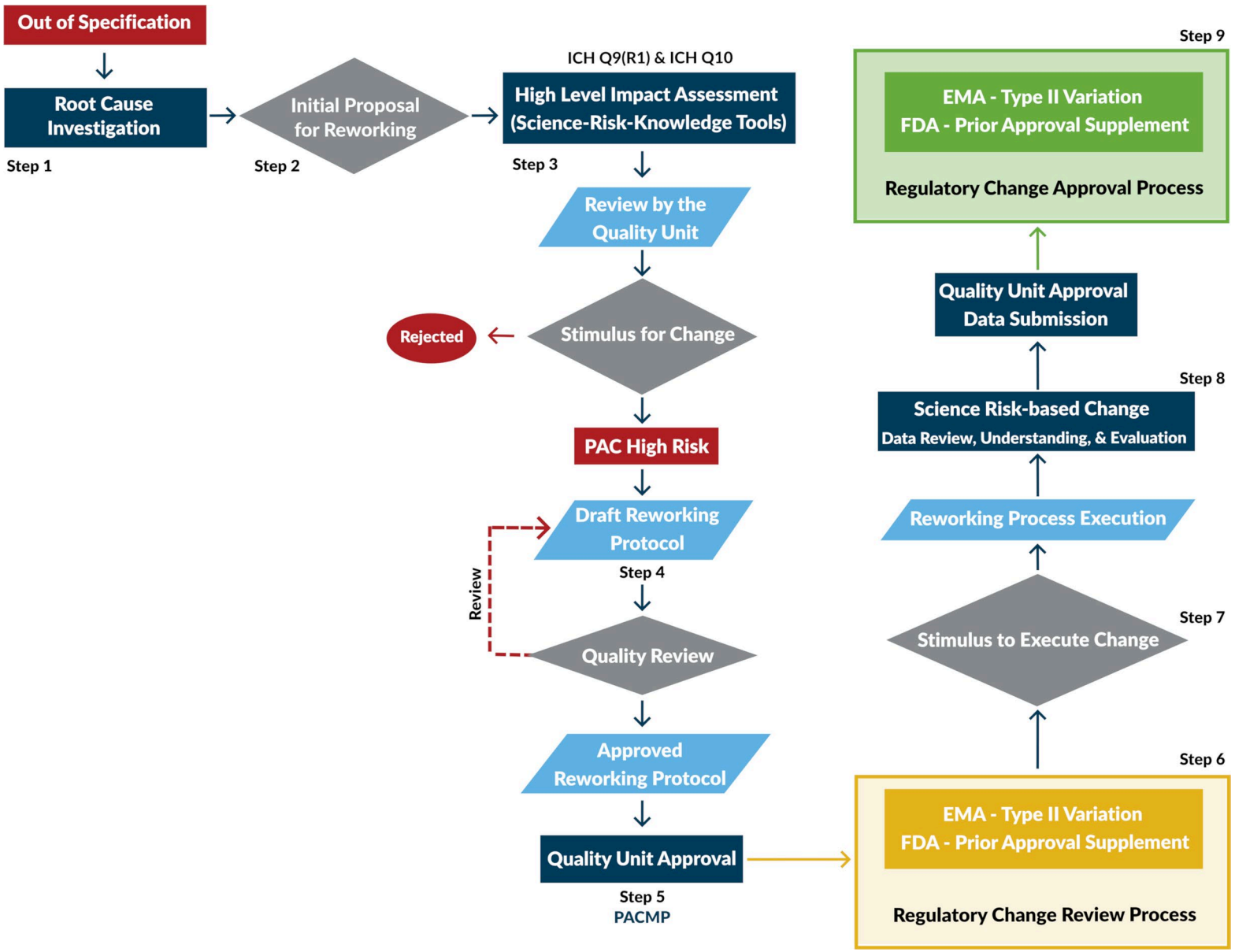

Figure 5.4-1 Proactive Reworking Approach

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5.5 Application of Knowledge and Risk Assessments

ICH *Quality Guideline Q10: Pharmaceutical Quality System* was the first official global guideline known to firmly position QRM and knowledge management (KM) as important enablers of an effective pharmaceutical quality system (PQS). Specifically, QRM and KM jointly enable (i) achieving product realization, (ii) establishing and maintaining a state of control, (iii) facilitating continual improvement, and (iv) critical decision-making during the reworking strategy ***(57)***.

ICH Q9(R1) emphasizes the relationship between risk and knowledge, and it highlights the importance of KM in managing risks. It places knowledge and KM repeatedly and explicitly at the core of all QRM activities, further reinforcing the connection of scientific knowledge and risk in several other regulatory guidance documents. Specifically, the definition of risk-based decision-making reads, "An approach to, or a process of, making decisions that considers knowledge about risks relevant to the decision and whether risks are at an acceptable level" ***(6)***.

The Risk Knowledge Infinity Cycle (RKI Cycle) establishes and promotes the following concepts:

- Knowledge is both an input to and an output from QRM
- Knowledge has an inverse relationship with risk
- Knowledge should flow effortlessly and on demand to inform risk and risk-based decisions
- The Cycle is continuous and perpetual

The RKI Cycle can be applied to support the goals of the PQS as defined in ICH Q10. A set of six principles corresponding with six "nodes" along the Cycle has been developed to better guide the tangible connection between QRM and KM practices as illustrated in **Figure 5.5-1** ***(58)***.

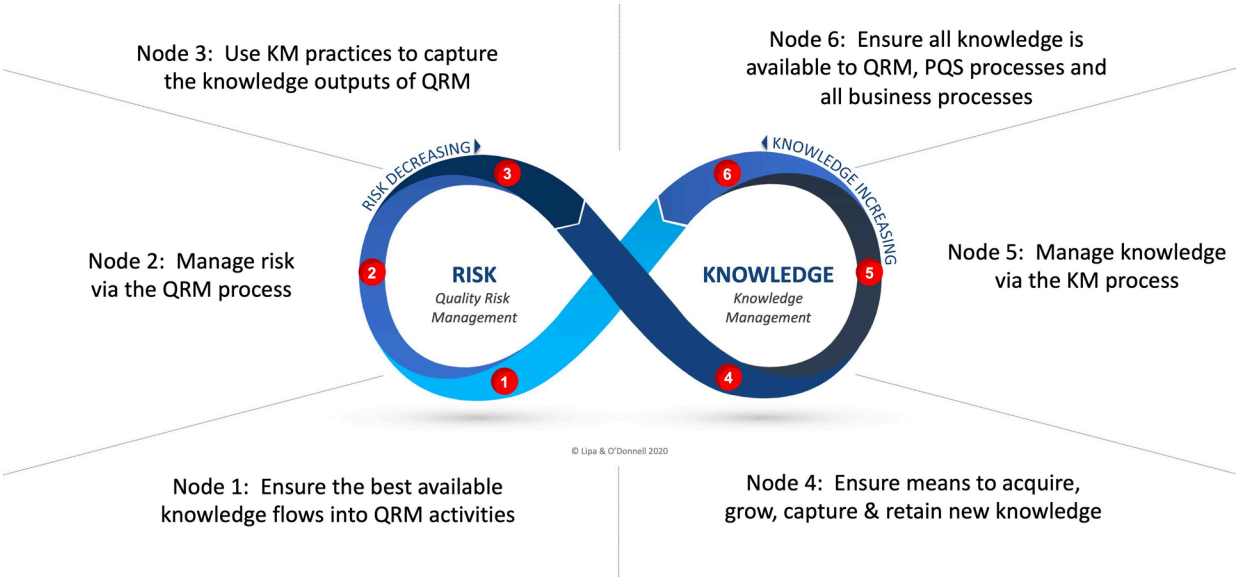

Figure 5.5-1 Six Guiding Principles of the Risk Knowledge Infinity Cycle Framework

During the reworking of biologicals, rigorous risk management ensures that any changes or corrective actions do not compromise critical quality attributes, safety, or regulatory compliance. Structured assessments (e.g., FMEA within a quality risk management framework) prioritize controls and justify decisions, while robust change control, deviation tracking, and CAPA link risks to measurable outcomes. Effective knowledge management captures batch histories, tacit operator knowhow, prior rework learnings,

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and real-time analytics in a shared repository. This enables cross-functional review and consistent, evidence-based decisions that reduce variability and prevent repeat issues.

Knowledge

Prior knowledge is a critical tool to identify and justify quality during the critical planning, decision-making, and execution of the OOS of the biological drug substance or a drug product to be reworked. Three forms of knowledge are critical:

1. Internal knowledge of the manufacturer gained throughout the product lifecycle (including development and characterization of biological drug substance and drug product) is the basis for identifying the elements of CMC that are established conditions (EC) ***(52)***, and those elements which are supportive information, historical development of products and/or similar processes, proprietary technologies, modelling data, previous regulatory filings, regulatory inspections, pre-approval manufacturing changes, compatibility assessments, CQAs and any signals associated with QSIP ***(4)***.
2. External knowledge comes from public databases, literature surveys, journals, conferences, peer-reviewed publications and monographs.
3. Knowledge application of scientific principles from biophysical sciences, pharmaceutical sciences, biochemical engineering, material science, and application of principles of engineering for evaluation of the structure-activity relationship of the molecule under investigation.

ICH Q9(R1) defines two primary principles of QRM:

4. The evaluation of the risk to quality should be based on scientific knowledge and ultimately linked to the protection of the patient.
5. The level of effort, formality, and documentation of the QRM process should be commensurate with the level of risk.

The details, in-depth information, and prior knowledge are critical during **Step 2** of **Table 5.4-1** and **Figure 5.4-1** (PAR). **Figure 5.5-1** provides an overview of the supporting information required to carry out a detailed assessment.

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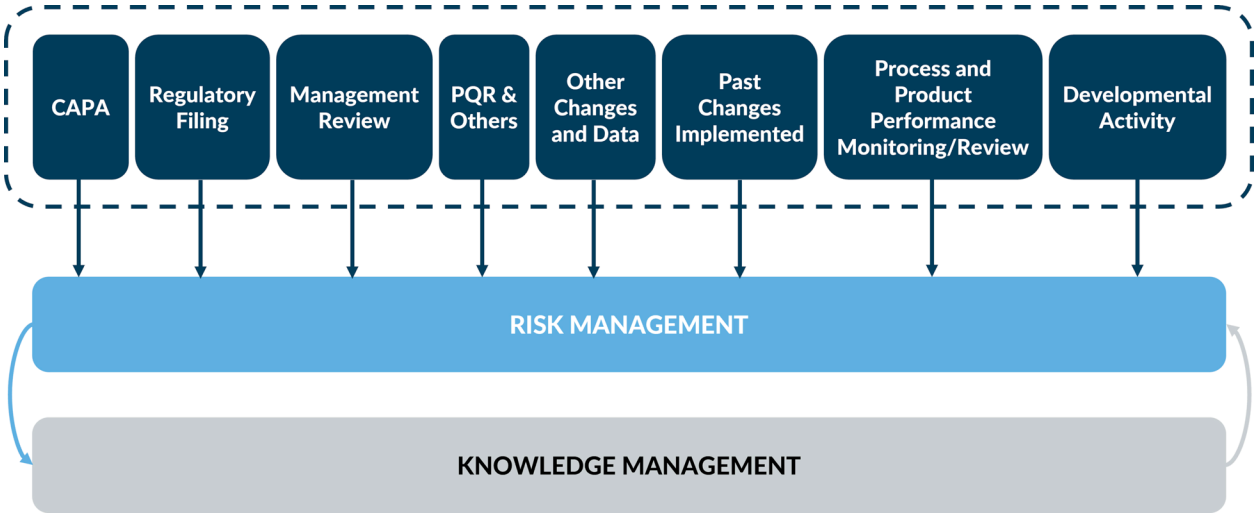

Figure 5.5-1 Information Required for Step 2 (Detailed Assessment)

5.6 Post-Approval Change Tools

When deciding to implement a PAC, thereby aiming for a new desired state, the tools suggested in **Table 5.6-1** can be beneficial while drafting the documentation and scientific justifications. The use of tools and points for consideration noted in **Table 5.6-1** can aid in identifying and presenting, at the outset, the most convincing and thorough documentation required for the approval process of Reworking. (See PDA *Technical Report No. 91: Post-Approval Change Management and Implementation for Biologics and Pharmaceutical Drug Products–A User's Guide.*)

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Table 5.6-1 Tools and Considerations for Decision-Making for Post-Approval Change *(49)*

	Defining the Scope of Change	Understanding	Risk Assessment	Regulatory Assessment	Definition	Approval Strategy	Implementation
Process	What is the scope/ background? What is rationale/ justification? What is the benefit? What other solutions have been evaluated?	Have ECs been established/filed? Have CQAs and CPPs been established? Has a control strategy been established? Has a PACMP been considered?	Determine categorization by performing a risk assessment of the change related to patient safety, product quality or efficacy If the risk is moderate to low → lesser filing may be considered If the risk is high → a prior approval submission is likely	Q12 PAC procedure or other procedure? Are ECs, CPPs, and/or CQAs impacted? If NO and has minimal impact → change may be handled in the PQS, specific jurisdiction notification may be required, impact to CTD needs to be assessed If YES → assess for specific filing type per jurisdiction	Define design space, as appropriate Defined performance space, as appropriate Define qualification/ validation activities, as applicable Create a stakeholder assessment– who might be affected and who has to be involved in the process?	Define approval for ICH countries Define approval strategy for countries following ICH guidelines Define approval strategy for all other countries Contact authorities according to strategy Prepare for implementation	Create/update PACMP, as required Submit protocol to authorities Implement change Verify implementation and validation Submit report to authorities
Tools	Business case Timeline System impact assessment	Product and process understanding Process risk assessment (QRM) Control strategy	Change risk assessment Quality risk assessment	PQS Change categorization	Quality by design Validation protocol Stakeholder assessment	Approval strategy Change categorization	PACMP Validation protocol and report

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A regulatory toolbox was jointly agreed upon by the EMA and the FDA on 26 November 2018. The agencies sought to identify scientific and regulatory solutions to challenges commonly experienced by applicants of early-access applications in completing Module 3 data requirements in time for the marketing authorization applications (MAA) ***(59)***.

5.7 Established Conditions

The implementation of ICH *Quality Guideline Q12: Technical and Regulatory Considerations for Pharmaceutical Product Lifecycle Management* by different regulatory agencies will also enable the recognition of ECs that require regulatory notification as well as those non-ECs that have been recognized as not necessary to ensure product quality and, therefore, can be managed prior to implementation notification ***(60)***. The amount of flexibility in a control strategy is based on the totality of product and process understanding (e.g., prior knowledge, development studies) in the context of QRM principles described in ICH Q9(R1).

A control strategy is designed to ensure that a product of required quality will be produced consistently. This planned set of controls, derived from current product and process understanding, assures process performance and product quality. The controls can include parameters and attributes related to drug substance and drug product materials and components, capability of the verified mathematical models, facility and equipment operating conditions, in-process controls, finished product specifications, and the associated methods and frequency of monitoring and control. The control strategy is based on the development of the CQAs which, in turn, have been based on thorough scientific work ***(4, 57)***.

Pharmaceutical development activities result in an appropriate control strategy of the ECs, which are critical elements identified and proposed by the MAH to assure product quality and, therefore, would require a regulatory submission if changes impacted any EC. ECs can include CPPs, critical material attributes (CMAs), and CQAs as defined in ICH Q8(R2) ***(4)***.

In the event of reworking of a drug substance or drug product, CMC changes can occur. Thus, the ECs for the manufacturing process should be defined based on enhanced product and process understanding, considering all the relevant elements of the control strategy. To identify the ECs, an enhanced approach is recommended with an increased understanding of interaction between the changed key and critical input parameters along with the outputs, and the impact on quality attributes together with a corresponding control strategy. Once ECs are identified for the material to be reworked, a detailed initial risk assessment must be developed from the prior knowledge gathered to identify the potential risk to product quality associated with changing the ECs and considering the overall control strategy.

5.8 Post-Approval Change Management Protocols

The PACMP tools described in ICH Q12 enable the planning and implementation of changes to ECs and/or CPPs, and thereby to regulatory filings, in an efficient structured manner. The PACMP (or comparability protocol) approved by the relevant health authority provides an agreement between the MAH and the regulatory agency and describes the changes and level of detail complying to the complexity, major variation, risk assessment, control strategy, and other pertinent elements. Once a PACMP is approved, when the proposed study design or acceptance criteria changes, an update of the protocol should be submitted for implementation as agreed upon in the original protocol. If changes are needed due to unanticipated or unforeseen changes in plans, prior approval of a new or revised PACMP by the regulatory agency may be required.

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In the event of a reworking, the proposed comparability protocol (CP) for a modified manufacturing process should sufficiently cover the intended change and provide an increased control when compared to the existing process ***(61)***. A CP can be appropriately designed and/or drafted using a risk-based approach, with an in-depth or a high-level knowledge of process and product understanding. A risk assessment should be conducted on the potential drug substance and/or intermediate CQAs to be affected by parameter changes. including unspecified parameters ***(12)***. Knowledge from previous developmental studies, small-scale studies, and/or pilot studies can be critical to identify potential risks to quality and provide support in the development of a suitable evaluation plan. The risk of adverse effects as a result of cumulative changes during a reworking should be carefully identified, justified, and included in the supporting information for the CP. The reworked material should be assessed with suitable structural characterization, analytical and procedures. It should meet the criteria to demonstrate the lack of adverse effect, including immunogenicity on the reworked product quality considering the impact of structural changes in drug substance and/or drug product. Depending on the type and complexity of the drug substance, functional characterization and additional studies should also be included.

5.9 Instabilities and Quality Concerns

Protein degradation ***(62),*** aggregation ***(18)***, poses significant challenges in the manufacturing and quality control of biologicals, particularly during reworking processes. Given that, generally, these proteins are only marginally stable in solution, they are susceptible to various forms of instability. Chemical instabilities like deamidation, oxidation, and hydrolysis, alongside such physical instabilities as denaturation and noncovalent aggregation, can compromise the quality and efficacy of the drug substances .

During an event of reworking, proteins will be exposed to stress conditions, such as elevated temperatures, low pH, or hydrophobic surfaces, which might trigger conformational instabilities ***(23, 63, 64)***. Formulation of drug substances at high concentrations can further encourage protein aggregation, impacting long-term stability ***(55)***.

To mitigate these risks, purification steps are essential for removing impurities and aggregates. It is crucial to implement appropriate process controls that manage pH, buffer components, and denaturants to prevent protein unfolding, which can lead to aggregation. These controls are vital to ensuring the drug substance meet specifications, maintains good appearance without visible particulates, retains stability during holding and storage, preserves efficacy, and minimizes immunogenicity concerns.

Protein manufacturing and formulation are complex processes with several factors that can impact the quality and stability of therapeutic proteins, such as:

- **Protein Concentration and Viscosity:** During reworking, increased protein concentration can lead to higher viscosity, which is problematic for formulations intended for subcutaneous injections. Techniques such as rheology and dynamic light-scattering can be employed to evaluate viscosity impacts and ensure optimal formulation characteristics.
- **Thermal Stress:** Elevated temperatures, particularly during viral inactivation steps, can lead to protein denaturation and aggregation. While it is challenging to quantitatively assess these impacts, careful control of temperature conditions is necessary to maintain protein stability.
- **High Ionic Strength:** The use of high-ionic-strength eluents in ion-exchange and hydrophobic interaction chromatography may promote aggregation. Monitoring and controlling ionic strength during these steps can minimize the risk.

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- **pH Considerations:** Eluents with pH values near the protein's isoelectric point, or far from it, can lead to aggregation. It is crucial to optimize and maintain pH conditions throughout processing to prevent destabilization.
- **Filtration Processes:** Membrane contact, presence of air bubbles, and shear stress during filtration can induce aggregation. Ultrafiltration, especially at high protein concentrations, requires careful management to prevent aggregation, particularly before filling bulk containers.
- **Formulation Challenges:** Bulk storage, freeze-thaw cycles, and agitation during formulation processes can stress proteins and cause aggregation. Implementing gentle handling techniques and optimized storage conditions can reduce these risks.
- **Air-Water Interfaces:** Exposure to air-water interfaces during formulation can promote aggregate formation by lowering the energy barrier for protein unfolding. Minimizing such exposure and conducting visual inspections can help identify and address potential aggregation issues.
- **Excipients in Formulation:** Certain excipients can stress proteins and accelerate aggregation. For instance, magnesium ions (Mg^{+2}) have been noted for their potential to increase aggregation rates in some formulations. Understanding and managing the impact of excipients is critical for maintaining biological molecule stability.
- **Visible Particulates:** It is critical to effectively monitor visible particulates after any reworking. If visible particles are detected, it is necessary to implement detection measures, conduct chemical identification, perform root cause analysis, and develop mitigation strategies for maintaining safety and efficacy of the reworked biologicals.
- **Leachables and Extractables:** Leachables and extractables typically pose minimal issues for traditional cellulose-based depth filters ***(33, 46)***. However, they have been detected in various sterilizing-grade filters, such as polyvinylidene fluoride, polyethersulfone, and mixed cellulose ester. These filters demonstrate a complex pattern of destabilizing and stabilizing effects on the physical and chemical stability of mAbs, including the formation of visible and subvisible particles. The leachables and extractables may also exhibit "surfactant activity." Characterized leachables and extractables in these filters include organic carbon, metals, inorganic salts, and fibers. Since protein solutions undergo multiple chromatographic and filtration steps, most extractables and leachables ***(65)*** should be removed during downstream processing.

By comprehensively addressing these areas through careful process controls, analytical techniques, and optimization strategies, the risks associated with protein aggregation can be significantly minimized, ensuring the production of high-quality therapeutic proteins. Understanding and controlling these factors are key to maintaining the integrity and quality of biologicals throughout the stages of reworking.

5.10 Roles and Communication

People drive reworking projects and ensure timely delivery of the biological drug product to the patient. Top management plays a critical role by setting the strategic direction and ensuring that all processes adhere to regulatory standards ***(57)***. Meanwhile, other team members can focus on operational aspects, ensuring each function, whether it is research, development, manufacturing, quality, logistics, or support, is executed efficiently.

5.10.1 Roles

Roles can be defined individually or integrated into teams, which allows for flexibility in management and execution, ultimately contributing to successful outcomes for the project and timely delivery to the patient ***(57)***.

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Their responsibilities include:

- **Providing Strategic Direction and Oversight**
  - Approving the rework plan, scope, and objectives
  - Ensuring alignment with organizational goals, regulatory requirements, and quality standards
- **Ensuring Regulatory Compliance and Risk Management**
  - Supporting adherence to GMP, regulatory guidelines, and internal SOPs
  - Overseeing risk assessments and mitigation strategies related to reworking activities
- **Allocating Resources and Budget**
  - Approving necessary budgets, personnel, facilities, and equipment for the rework process
  - Ensuring that sufficient resources are available to execute the rework effectively
- **Facilitating Cross-Functional Collaboration**
  - Promoting communication among manufacturing, quality assurance, regulatory, and other relevant departments
  - Ensuring all stakeholders are informed and involved in decision-making
  - Departments should explicitly include contract manufacturers responsible for intermediates and supply chain/demand planning teams, and manufacturing resilience given their impact on production scheduling, inventory, and communication with commercial functions
- **Monitoring Progress and Performance**
  - Receiving regular updates on the status, challenges, and outcomes of the rework process
  - Making high-level decisions to address issues or deviations promptly
- **Ensuring Regulatory Submissions and Documentation**
  - Approving necessary documentation for regulatory filings, deviations, or change controls associated with reworking
  - Ensuring that all activities are documented thoroughly for compliance and traceability
- **Managing Ethical and Safety Considerations**
  - Prioritizing patient safety and product quality throughout the process
  - Supporting a culture of transparency and accountability
- **Post-Rework Evaluation and Continuous Improvement**
  - Reviewing rework outcomes, including quality, stability, and regulatory feedback

Implementing lessons learned to improve future processes. In essence, top management plays a strategic and oversight role, ensuring that the reworking process is carried out responsibly, efficiently, and in compliance with all applicable standards, thereby minimizing risk to the patient, safeguarding product quality and organizational integrity.

5.10.2 Communication

Effective communication is essential to ensure that all stakeholders are informed, coordinated, and aligned throughout the reworking process ***(57)***. The best approach involves a structured, transparent, and timely communication strategy that minimizes misunderstandings and supports compliance. The key communication principles and steps are:

- **Develop a Communication Plan**
  - Define objectives: Clarify what needs to be communicated, including process updates, risks, decisions, and regulatory requirements.

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  - Identify stakeholders: Build internal teams (Quality, Manufacturing, R&D, Regulatory Affairs, Management) and external parties (regulators, partners, suppliers).
  - Establish channels: Use formal channels such as meetings, memos, reports, and digital platforms to ensure traceability and clarity.
  - Set frequency: Determine how often updates are needed—daily, weekly, or as significant events occur.
- **Ensure Transparency and Clarity**
  - Clearly articulate the reasons for rework, potential impacts, and actions being taken.
  - Communicate known risks and mitigation strategies upfront.
  - Provide timely updates on progress, issues encountered, and resolution plans.
- **Maintain Documentation and Traceability**
  - Record all communications, decisions, and approvals related to the rework.
  - Use controlled documents and logs to ensure information integrity and facilitate audits.
- **Facilitate Cross-Functional Collaboration**
  - Hold regular cross-disciplinary meetings to discuss progress, challenges, and decisions.
  - Encourage open dialogue to surface concerns or uncertainties early.
- **Implement Regulatory Communication Protocols**
  - If rework impacts product quality or regulatory filings, prepare necessary documentation for regulators.
  - Communicate with regulatory bodies as per established procedures, providing updates and seeking approvals if required.
- **Designate Clear Roles and Responsibilities**
  - Assign communication responsibilities to specific personnel (e.g., project manager, Quality head, process science/MSAT head).
  - Ensure all team members know to whom to report and how to escalate issues.
- **Use Clear, Concise, and Consistent Messaging**
  - Avoid technical jargon when communicating with nonexperts.
  - Ensure messages are consistent across all channels to prevent confusion.
- **Prepare for Crisis or Unexpected Events**
  - Develop contingency communication plans for issues such as contamination, delays, or regulatory findings.
  - Keep stakeholders informed promptly to manage expectations and facilitate swift responses.

The best approach to communication management during reworking is a proactive, transparent, and well-structured strategy that involves developing a detailed communication plan, ensuring cross-functional collaboration, maintaining thorough documentation, and engaging regulatory authorities when needed. This approach fosters trust, minimizes risks, and ensures compliance throughout the reworking process.

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5.11 Submission & Approval Timelines

Reworking a biological drug substance involves carefully revisiting and potentially modifying production processes to ensure product quality, safety, and efficacy. The timeline for reworking such a drug substance and/or drug product can vary significantly depending on several factors, including the complexity of the product, extent of modifications needed, regulatory requirements, and manufacturing facility's capabilities. Below are typical phases and approximate timelines:

1. **Assessment and Planning (2–4 weeks)**
   - Evaluation of the current process and product
   - Identifying reasons for rework
   - Developing a detailed rework plan and risk assessment
2. **Process Development and Optimization (4–12 weeks)**
   - Adjusting upstream or downstream processes
   - Small-scale laboratory or pilot studies to validate modifications
   - Analytical method development or validation if needed
3. **Analytical and Stability Testing (4–8 weeks)**
   - Confirming product integrity, potency, and purity post-rework
   - Stability testing to ensure product remains within specifications
4. **Scale-up and Validation (8–16 weeks)**
   - Manufacturing at larger scale
   - Process validation batches to confirm reproducibility
5. **Regulatory Documentation and Submission (4–12 weeks)**
   - Preparing amendments or supplemental dossiers for regulatory authorities
   - Addressing agency queries and obtaining prior approval
6. **Manufacturing and Release (variable, typically 2–6 weeks)**
   - Final production run
   - Quality release-testing and documentation

Total Estimated Timeline

Typically, reworking a drug substance can take approximately 6 to 12 months, but in some cases, especially with complex modifications or regulatory hurdles, it may extend longer.

Factors Influencing Reworking Timeline

Several crucial factors that influence the timeline for reworking a drug substance include:

- **Regulatory Pathways:** The need for prior approval or additional filings can significantly affect timelines. Approval processes differ by region and can vary based on the scope of modifications.
- **Extent of Process Changes:** More extensive changes typically require deeper assessments and validation efforts, which lengthens the timeline.
- **Availability of Raw Materials and Manufacturing Capacity:** Supply chain considerations and manufacturing capacity can cause delays, particularly if specific raw materials are scarce,

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facility constraints exist, or there are changes in any equipment or introduction of single-use systems.

- **Quality and Analytical Testing Turnaround Times:** The time it takes to conduct necessary quality and analytical testing can impact timelines, especially if retesting is required or if results necessitate further adjustments.
- **Single-use Systems:** If new single-use systems with greater process contact-surface area or new polymers are introduced that encounter the drug substance during bulk dispensing containing excipient buffers or final filtration a detailed leachable and extractable impact assessment will need to be performed before execution of reworking process ***(65-67)***.
- **Regulatory Agency Review Times:** Variability in how long agencies take to review submissions and respond can introduce uncertainties in the timeline.

Starting from the point of prioritization—the main reason to rework a drug substance t—the timeline for reworking is highly variable and should be planned carefully, incorporating all development, validation, and regulatory steps to ensure compliance and product quality. Consulting with regulatory authorities early in the process can help streamline the timeline and clarify requirements.

5.12 Quality Unit Preparedness

The Quality Unit requires comprehensive training documents to ensure compliance and efficiency during unexpected events of reworking ***(57)***. Here are the typical documents that might be needed:

- **Standard Operating Procedures (SOPs):**
  - SOPs for rework processes that include step-by-step instructions.
  - SOPs for identifying and segregating nonconforming materials.
- **Regulatory Guidelines:**
  - Relevant regulatory guidelines from authorities like FDA, EMA, or other local or national standards related to rework.
- **Quality Control Procedures:**
  - Documents detailing the quality control checks required before, during, and after rework.
- **Safety Training Documents:**
  - Training materials on safety procedures specific to rework operations.
- **Documentation and Recordkeeping Training:**
  - Instructions for accurate documentation and recordkeeping during rework to ensure traceability and compliance.
- **Risk Assessment and Management:**
  - Guidelines for conducting risk assessments related to reworking processes.
- **Material Handling Procedures:**
  - Proper procedures for handling materials safely during the rework process.
- **Equipment Handling and Maintenance:**
  - Training on the specific equipment used for reworking, including maintenance and calibration.
- **Continuous Improvement Practices:**
  - Maintain documents that facilitate identifying areas for improvement in rework processes.

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To effectively prepare for a reworking event, the Quality Unit can follow these steps:

1. **Risk Assessment and Planning:**
   - Conduct a thorough risk assessment to identify potential issues and their impact on product quality.
   - Prepare a detailed rework plan, including objectives, necessary resources, timelines, and personnel responsibilities.
2. **Documentation and SOPs:**
   - Ensure all SOPs related to rework are up to date.
   - Document every step of the rework process clearly and maintain traceability.
3. **Training:**
   - Provide training for personnel involved in reworking activities on new procedures, use of equipment, and quality requirements.
4. **Material Management:**
   - Review and inspect raw materials or components and consumables involved in rework to ensure they meet quality standards.
   - Establish a system for the segregation of nonconforming materials.
5. **Monitoring, Review, and Control:**
   - Set up monitoring and control mechanisms to ensure consistent quality during the rework process.
   - Implement checks and balances to detect any deviations promptly.
6. **Communication:**
   - Maintain clear communication channels within the team and with other departments involved.
   - Keep stakeholders informed of progress and any issues that arise.
7. **Quality Assurance:**
   - Conduct quality assurance checks before, during, and after the rework process to ensure compliance with the original specifications.
   - The quality unit is ultimately responsible for ensuring product quality of any lot or batch undergoing reworking activities and for providing quality oversight of the manufacturing process throughout any reworking activity
8. **Feedback and Improvement:**
   - After completing the rework, gather feedback and conduct a thorough review of the process to identify areas for improvement.
   - Update policies and procedures to reflect lessons learned.

It is crucial for the quality unit, comprising both quality assurance and quality control, to be well-prepared to manage the situations effectively and ensure that product quality and compliance are maintained. Ultimately, preparedness involves a combination of detailed planning, ongoing training, and resource management and ensuring that the team is proactive in adapting to any new challenges that arise during the reworking process.

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5.13 Conclusions

In urgent situations, such as a global pandemic (e.g., COVID-19), wars, or natural disasters (e.g., Category 5 hurricanes, catastrophic earthquakes, major floods, meteor catastrophe), certain regulatory flexibilities or deviations may be considered by the MAH to address critical supply issues, like immediate drug shortages. However, reworking an OOS drug substance generally involves significant quality and regulatory considerations, such as:

- **Regulatory Frameworks and Guidance:**
  - Regulatory agencies such as the U.S. FDA, EMA, and other agencies, offer guidance on managing deviations, OOS results, and manufacturing challenges. In urgent situations, these agencies may provide tailored guidance or emergency procedures. However, deviations from approved specifications generally require proper justification, and thorough documentation.
- **Reworking of OOS Drug Substance:**
  - Reworking involves processing material that initially failed to meet specifications to bring it into compliance.
  - This process must be validated and scientifically justified, demonstrating that reworking does not compromise product quality, safety, stability and efficacy.
- **Regulatory Approval and Documentation:**
  - Any decision to rework should be thoroughly documented, including risk assessments and validation data.
  - For licensed products, regulatory authorities often require that deviations or exceptions from established processes be reported. Depending on the jurisdiction and the specific nature and impact of the change, prior approval may be necessary. This ensures compliance with regulations and maintains product quality and safety standards. It is crucial to follow the guidelines and procedures set forth by relevant regulatory bodies to avoid potential implications on the product's marketing authorization.
- **Risk Management and Justification:**
  - In emergencies or shortages, regulators may offer some flexibility, but the company must ensure reworking does not compromise drug quality.
  - A comprehensive risk assessment, incorporating knowledge management, should be conducted to evaluate its impact on safety and efficacy.
- **Good Manufacturing Practice Compliance:**
  - Reworking must adhere to GMP standards, and the process should be validated.
  - Proper change control procedures should be followed.

Reworking an OOS drug substance should never be treated as routine. However, in urgent situations such as a pandemic or critical shortage, biotechnology companies may consider reworking if it is properly justified, validated, and documented. They should proactively engage with regulatory authorities, possibly under emergency use or contingency provisions, to ensure compliance and maintain product integrity.

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6.0 Appendix I: Hypothetical Case Study (Reprocessing Framework) — Anion Exchange Chromatography)

Section 6.0 (Appendix I) focuses on the design and implementation considerations for a reprocessing step. Adapted from the published monoclonal antibody manufacturing (mAb) process, *A-Mab: A Case Study in Bioprocess Development,* for this case study, a reprocessing framework has been created using an anion exchange chromatography (AEX) step ***(19)***. This reprocessing framework is intentionally designed to require a more comprehensive analysis than a simpler reprocessing step, such as refiltration described in **Section 7.0**.

Reprocessing of a major unit operation, such as an AEX chromatography column, requires a detailed assessment, significant resources, planning, and commitment. Simpler unit operations require a subset of the planning and activities outlined here. Regardless, quality management systems, regulatory requirements, and a comprehensive process understanding should drive the overall design of the reprocessing procedure.

6.1 Unit Operation Characteristics

Chromatography is used to purify a product (e.g., mAb, target proteins, compounds) considering the properties of the target solute, feed stream, required purity level, productivity, costs, process sequence, process parameters, and implementation of the designed purification process. Biomolecules are purified using chromatography techniques that separate them according to the difference in their specific properties as summarized in **Table 6.1-1.** In ion exchange chromatography (IEX), biomolecules are separated according to the difference in their net surface charges.

Table 6.1-1 Chromatography Techniques

Property	Techniques
Charge	Ion exchange chromatography
Size	Size exclusion chromatography, also called gel filtration
Hydrophobicity	Hydrophobic interaction chromatography, reversed phase chromatography
Biorecognition (ligand specificity)	Affinity chromatography

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The overall goal of a chromatographic step is to remove impurities, such as host-cell-derived proteins (HCP), nucleic acids, carbohydrates, and other undesired compounds.

In general, a full-scale chromatography system will consist of the following components:

- **Buffer and Product Reservoirs:** Containers (hard-piped or single use disposables) used to supply the buffers, product feed, and for elution collection required during the chromatography operation.
- **Pumps:** Peristaltic pumps used to feed the buffers into chromatography columns. Depending on scale, other types of pumps may be utilized considering process requirements and equipment design.
- **Column:** The chamber where the chromatographic media is loaded. The size (diameter and height) of the column will be defined based on the process requirements. The material of construction (e.g., glass, polycarbonate, steel) is selected based on process conditions (e.g., pressure and flow rates).
- **Resin:** Chromatography media intended for the purification process. The packaging of the columns is commonly conducted by creating a slurry where the particles are suspended, that is, poured into the column.
- **Valves:** Sanitary systems used to control the flow of materials during the operation of the chromatography system. These may be manual or pneumatic units.
- **Sensors:** Applications intended to monitor process parameters required for the chromatography process. These may include pH, air, temperature, flow, pressure, and UV monitor.
- **Controller/Computer:** Used to control and monitor the operation of the system.

The purification strategy involving a chromatography step is based on the characteristics (e.g., properties and stability) of the product under different conditions, final specifications, feed properties (e.g., concentration of product and impurities), nature of the process (e.g., continuous or discontinuous), and nature of the substances (e.g., hazardous). When designing the reprocessing step for the AEX chromatography case study, consideration was given to the quality attributes for mAbs listed in **Table 6.1-2** ***(19).***

Table 6.1-2 Summary of Quality Attribute Risk Assessment Topics

Product Quality Attribute		
Aggregation*	High mannose content*	Host-cell proteins*
Galactose content*	Nonglycosylated heavy chain*	Sialic acid content*
Afucosylation* Oxidation	C-terminal lysine	DNA
Protein A	Deamidated isoforms	Methotrexate
*Considered CQA		

6.2 Points to Consider

The elements in **Section 6.2** should be considered as part of reprocessing a chromatography column. These elements are further discussed in PDA *Technical Report No. 14 (Revised 2008): Validation of Column-Based Chromatography Processes for the Purification of Proteins*.

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6.2.1 Column Packing

The resin utilized for chromatography is loaded into the column in a controlled manner to ensure a uniform distribution while preventing the formation of channels that will affect the performance of the unit operation. Commonly, a slurry containing resin in a process buffer is poured into the column. Once the resin is settled, the integrity of the packaged column is confirmed. In the case of reprocessing, the integrity of the column bed should be verified and confirmed prior to its use. Any issue affecting the bed should be solved prior to reprocessing, considering resuspension and repacking of the resin if applicable.

6.2.2 Cleaning and Sanitization

Full cleaning and sanitization of the column must be conducted in preparation for reprocessing. Cleaning and sanitization will follow existing procedures and practices, including the regeneration of the resin. (These elements are discussed further in ***(68, 69)***.)

6.2.3 Performance

Performance of the chromatography step must be monitored according to the routine established procedures and must satisfy predefined acceptance criteria. The criteria should include, but not be limited to, the operational conditions (e.g., asymmetry, height equivalent to the theoretical plate, and/or transition analysis) and quality attributes ***(65)***.

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6.2.4 Regeneration and Lifecycle

The resin must be regenerated consistent with established procedures and must satisfy the acceptance criteria defined and validated. The use of the resin for a reprocessing activity will be within the validated number of reuses if the reprocessing activity is not associated with a malfunction of the chromatography column and resin.

6.2.5 Validation and Filings

Validation of reprocessing will be performed under a prospective protocol. Detailed guidance for validation activities of chromatography processes for the purification of proteins is provided in PDA TR 14 ***(68)***. In the event that changes are required to an existing filing, guidance is provided in PDA *Technical Report No. 38: Manufacturing Chromatography Systems Post-Approval Changes (ChromPAC): Chemistry, Manufacturing, and Controls Documentation.*

In this example, a protocol for prospective validation of reprocessing at the manufacturing scale was written and approved; however, the potential of a concurrent validation could be considered if the frequency for reprocessing did not justify a prospective protocol. The protocol required a single GMP run at manufacturing scale, including all routine in-process and release testing as well as additional nonroutine testing, for example, analysis of HCP, acidic variants, aggregate levels, fragmentation, oxidation, and potency in reprocessed material. The protocol also required placing the resulting drug substance in the stability program. The protocol acceptance criteria for nonroutine testing were based on the comparability criteria established in **Section 6.4.3** and demonstrated at small-scale. (See **Section 6.5.6**.)

Yet, a full prospective validation approach may not be feasible or practical for reprocessing in all cases. Instead, a concurrent validation strategy may be more appropriate, allowing the manufacture and testing of products and only reprocessing when and if necessary. In the case of concurrent validation, the acceptance criteria shall be aligned with the approved specifications for the process.

The initial BLA to the FDA and MAA to the EMA for this new biologic included a complete manufacturing process development section for the AEX unit operation within Section 3.2.S.2.6 of the CTD. A comparability protocol was provided post-licensure to gain agreement with FDA on the implementation strategy and to request a reduction in reporting requirements from a PAS to a CBE-30 (i.e., documentation of changes being effected, 30-day response from FDA required). In the EU MAA, a Change Management Protocol was included that also outlined the implementation strategy to seek a reduction in reporting requirements from a Type II Variation to a Type IB Variation upon successful completion of the protocol. **Table 6.2.5-1** details the content required to support this type of regulatory strategy. The approach described is primarily focused on the U.S. and EU regulatory authorities that have defined regulatory pathways for this type of submission; however, the information described is also useful for regulatory filings in other countries.

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Table 6.2.5-1 Regulatory Submission Content

Common Technical Document (CTD) Section	Content	Reference to Specific Section
3.2.S.2.6	For each step that will include reprocessing, provide the following:	
	Identify in-process tests or manufacturing data that will be used to detect the failure	6.5.2
	Summarize the design of the unit operation including step purpose	6.5.3
	Provide risk analysis of the unit operation in detail and demonstrate that reprocessing would not negatively impact CQAs	6.5.4 6.4.1 6.5.5
3.2.R	Provide a detailed Comparability Protocol FDA (BLA) or Change Management Protocol (EU MAA) that seeks implementation of the reprocessing operation with reduced regulatory reporting category post-licensure and include the following information:	
	Provide a purpose statement indicating in which situations reprocessing may be used	6.4.3
	Provide a high-level description of the proposed reprocessing operation	6.5.2
	List any additional in-process controls or extensions to intermediate hold times	6.5.3
	Outline the strategy for process validation (e.g., single confirmatory batch at production scale)	6.5.4 6.5.8 6.2.5
	Provide the tests and acceptance criteria that will be used to evaluate the impact on CQAs or provide a cross-reference to appropriate content in other sections (e.g., 3.2.S.4.1, 3.2.S.2.4)	6.2.5 6.4.1 6.4.5

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Common Technical Document (CTD) Section	Content	Reference to Specific Section
3.2.R	Include a specific reference to the section in 3.2.S.2.6 that contains the risk assessment and laboratory-scale data already obtained for reprocessed batches	6.4.1 6.5.5 6.4.2 6.5.1
	Include an evaluation of the effect of the reprocessing step on the subsequent virus removal where applicable (3.2.A.2 Adventitious Agents)	6.5.4 6.4.1 6.5.5 6.5.8 6.5.9
	Make a commitment to provide the data for the confirmatory production-scale batch in a CBE-30/Type IB Variation, including the following:	6.5.8
	3.2.S.2.5 Process Validation	6.2.5 6.4.5
	3.2.S.4.4 Batch Analysis	6.2.5 6.4.5
	3.2.S.7 Stability (part of the AR)	6.4.3

6.3 Scope of Reprocessing

6.3.1 Determining the Need for Reprocessing

Table 6.3.1-1 describes the scoring criteria for prospectively assessing the potential need for reprocessing in the hypothetical A-Mab process using an FMEA approach. The FMEA assesses the severity and detection of CQAs and how readily they are detected during unit operations. Scores are determined for **Sections 6.5.2-6.5.5** and **Sections 6.4.1-6.4.3** according to the criteria below.

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Table 6.3.1-1 FMEA Scoring Criteria

Scoring Category	Score		
	1	3	5
Severity (S): Scored based on the potential of the failure to adversely affect a critical quality attribute	Will not affect a CQA	May lead to an effect on a CQA	Will lead to an effect on a CQA, possibly resulting in CQA failure
Occurrence (O): Scored based on the probability of the failure mode occurring	1 in >1000 batches	1 in 100 to 1000 batches	1 in <100 batches
Detection (D): How readily the failure mode could be detected prior to completion of the step	Can readily detect	Indirect ability to detect	Cannot readily detect

The FMEA risk assessment for the A-Mab manufacturing process is illustrated in **Table 6.3.1-2** and only concerns the purification steps.

The AEX chromatography step was operated in the flow-through mode, binding impurities such as DNA, HCP, endotoxin, and viruses to the resin while the antibody passed through. In addition, this step was designed to reduce levels of aggregate and acidic variants. Within the operating parameters of the AEX step, in this example, high levels of HCP in the column effluent are possible on rare occasions, and therefore, HCP is the quality attribute most likely to trigger reprocessing.

The risk assessment identified the AEX column (**Table 6.3.1-2)** as the step with the highest RPN. Development data indicated that the risk of high HCP levels was an inherent process risk and a rare event attributable to multiple, interacting sources of variability in the unit operation. Thus, the conclusion was that the AEX unit operation was a candidate for reprocessing.

The assigned scores are multiplied together (S x O x D = RPN) to determine the risk priority number (RPN). For example, for the "anion exchange chromatography" reprocessing step in **Table 6.3.1-2**, S x O x D = 5 x 5 x 5 = 125.

Note: For HCP limits in **Table 6.3.1-2**: an appropriate IPC limit for DS HCP would typically be set at 70–80% of the action limit (approximately 70–80 ng/mg; a practical choice is 80 ng/mg). This provides a buffer to account for assay variability and process drift, enabling corrective actions before reaching the <100 ng/mg action limit.

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Table 6.3.1-2 Application of FMEA to the AEX Reprocessing Step (Example 1)

Reprocessing Step	S	Comments	O	Comments	D	Comments	RPN
Protein A Chromatography	3	HCP may be affected by loading and elution pH.	1	Based upon manufacturing experience, pH or loading excursions outside the normal operating range (NOR) may occur in 1 of >1000 batches.	1	Automated process control skids and batch record checks would readily detect a CPP excursion.	3
Low pH inactivation	5	Aggregation and viral inactivation may be affected by the pH, temperature, and time.	1	Based upon manufacturing experience, pH or loading excursions outside the NOR may occur in 1 of >1000 batches.	1	Batch record checks would readily detect a CPP excursion.	5
Cation Exchange Chromatography	5	Aggregate and HCP may be effected by loading, load/ wash conductivity, elution pH, and elution collection.	3	Based upon manufacturing experience, excursions outside the NOR may occur in 1 of 100-1000 batches.	1	Automated process control skids and batch record checks would readily detect a CPP excursion.	15
Anion Exchange Chromatography	5	HCP removal may be impacted by equilibrium/ wash conductivity and load pH within the NOR. High conductivity and low pH are maximum challenge conditions.	5	Variability in this chromatography step occasionally leads to reduced HCP removal in 1 in <100 batches.	5	Failing levels of HCP at this step would not be detected.	125

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Reprocessing Step	S	Comments	O	Comments	D	Comments	RPN
Small-virus Retentive Filtration	5	An integrity failure of the virus retentive filter would impact the viral safety claim.	5	Based upon manufacturing experience, integrity test failures will occur in 1 of 50 virus filters.	1	Integrity testing of the virus retentive filters will identify a failure mode.	25
Small-virus Retentive Filtration	5	Exceeding a CPP limit would impact the viral safety claim.	3	Based upon manufacturing experience, overloading errors will occur in 1 of 100 batches.	1	Automated process control skids would readily detect a CPP excursion.	15
Formulation: Ultrafiltration/ Diafiltration	5	Product concentration will be affected by drug substance final dilution. Reconcentration may be required.	3	Based upon manufacturing experience, dilution errors will occur in 1 of 100 batches.	1	Concentration testing of the preformulated bulk will identify a failure mode.	15

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6.3.2 Business Case and Project Charter

Three options were considered in assessing the business case for reprocessing at the AEX step:

1. Discarding the batch upon failure to meet the HCP in-process control requirement of ≤80 ng/mg
2. Redesigning the AEX step or adding additional steps
3. Reprocessing during the AEX step

From a business perspective, it was confirmed that all three options could be managed to ensure acceptable product quality at release, but Option 3 was the most cost-effective and could be designed to meet quality and regulatory requirements. A project charter was created, and a cross-functional team was formed to develop a reprocessing step.

6.4 Quality Considerations

6.4.1 Quality Attribute Impact Assessment

A risk analysis was performed to assess the effect of repeating the AEX chromatography step on the quality attributes (**Table 6.4.1-1**). The potential effect of the reprocessing step on both drug substance and intermediate quality attributes was assessed. For all quality attributes listed, it is assumed that the company has previously developed analytical techniques able to detect each, resulting in Detectability scores of 1. Attributes impacted by the AEX step were identified and included in **Table 6.4.1-1.** Attributes with an RPN ≥15 were identified for further evaluation.

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Table 6.4.1-1 Application of FMEA to the AEX Reprocessing Step (Example 2)

	FMEA Scoring Process Parameter or Other Consideration	S	O	D	RPN	Comments (S, O, D)
Output AEX-related Quality Attributes	Protein concentration 7-8 g/L	3	5	1	15	Quality attribute, reprocessing will affect eluate concentration, detectable
	pH 7.4-7.6	1	1	1	1	Quality attribute, reprocessing will not affect pH, detectable
	Aggregate <0.8%	5	3	1	15	Quality attribute, reprocessing may affect eluate aggregate, detectable
	Acidic variants ≈10%	5	3	1	15	Quality attribute, reprocessing may affect eluate acidic variants, detectable
	Fragmentation	5	3	1	15	Quality attribute, reprocessing may affect fragmentation, detectable
	Potency	5	3	1	15	Quality attribute, reprocessing may affect potency, detectable
	Oxidation	5	3	1	15	Quality attribute, reprocessing may affect oxidation, detectable
	HCP <12 ng/mg	5	3	1	15	Quality attribute, reprocessing may affect HCP, detectable
Other	Leachables and extractables	3	3	3	27	Exposure of the API to the resin/filters a second time may increase potential extractables and leachables load
	Buffer/solution supply	5	5	1	25	Manufacturing may not have sufficient buffer to execute reprocessing; buffer-generation timing and impact on hold time need to be considered
	Filter sizing	3	3	1	9	Risk of aggregate formation during ambient storage is a concern; need to determine sufficient filtration area
	Equipment cleaning	5	5	1	25	Equipment cleaning turnaround time and impact on hold time need to be considered

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6.4.2 Analytical Methods

In addition to routine drug substance release-testing, additional nonroutine analytical tests were identified as necessary for the development and implementation of reprocessing. In this case, an analysis of HCP, acidic variants, aggregate levels, fragmentation, oxidation, and potency in reprocessed material was performed during reduced-scale development studies. All of these characterization assays were either validated or qualified as appropriate for use with samples from the AEX eluate.

The same analytical methods performed during the laboratory-scale models were planned for subsequent full-scale verification runs (comparability protocol). In addition to routine release and stability-testing, the same nonroutine testing performed on AEX eluate in laboratory-scale models would be performed at manufacturing scale.

Table 6.4.2-1 Summary of Analytical Methods to Assess the Product Quality Attribute

Analytical Methods to Assess the Product Quality Attribute	
Attribute	**Analytical Test**
Aggregates	Size-exclusion chromatography (SEC) Reverse-phase high-performance liquid chromatography (RP-HPLC) Fluorimetry
HCP	Liquid chromatography-mass spectrometry (LC-MS) Enzyme-linked immunosorbent assay
Fragments	SEC RP-HPLC Isoelectric focusing
High Molecular Species	SEC
Low Molecular Species	Capillary electrophoresis-sodium dodecyl sulfate
Acidic variants	IEX Capillary isoelectric focusing Imaged capillary isoelectric focusing
Galactose content	HPLC
Afucosylation	LC-MS
Oxidation	Peptide mass fingerprinting
C-terminal lysine	LC-MS

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Analytical Methods to Assess the Product Quality Attribute	
Attribute	**Analytical Test**
Deamidated isoforms	Capillary zone electrophoresis (CZE) Mass spectroscopy
Sialic acid content	HPLC-based fluorometric assay
Methotrexate	HPLC Liquid chromatography/tandem mass spectrometry (LC-MS/MS) UV–Vis spectrophotometry
Subvisible particles	Light obscuration method
Visible particles	Visual method Micro-flow imaging Dynamic light scattering
Unfolding	Differential scanning fluorimetry Protein conformational array
Disulfide Bonds	Gel electrophoresis (SDS-PAGE) Capillary electrophoresis (CE-SDS)

6.4.3 Comparability Criteria

For laboratory-scale models, the intermediate quality attributes had to remain consistent with AEX eluate that had not been reprocessed, with the exception of HCP, which was expected to be reduced by the reprocessing event.

For manufacturing-scale studies, all drug substance release specifications continued to be met. Additionally, the intermediate quality attributes remained consistent with AEX eluate that had not been reprocessed, with the exception of HCP, which was expected to be reduced by the reprocessing event.

Drug substance from the first reprocessed batch was placed on the established real-time and accelerated stability protocols. It was determined that evaluation of the resulting drug product stability was not required.

6.4.4 Monitor Triggers

The company continued to monitor the HCP in-process control test (action limit ≤100 ng/mg) at AEX eluate. (See **Section 6.5.2**.)

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6.4.5 Investigate, Execute Protocols, Analyze Data, Document Results, and Complete Commitments

Should the trigger described in **Section 6.4.4** be reached, the event would prompt the company to investigate, execute protocols, analyze data, document results, and complete commitments. In all cases where reprocessing occurred, an investigation would be conducted evaluating the root cause, product quality impact, and corrective actions.

6.5 Process Considerations

6.5.1 Development History

The development history of the AEX-column step should be reviewed to understand the purpose(s) of the original unit operation and the proposed reprocessing step. Control of CQAs and the potential impact to subsequent processing steps should be examined. Development studies that should be repeated or augmented to support reprocessing should be identified and discussed as described in **Sections 6.5.2-6.5.4**.

When considering a step for reprocessing, the original criticality assessment (**Table 6.5.1-1**) and an understanding of the original design parameters (**Table 6.5.1-2**) served as the technical basis for the design of the reprocessing procedure described in **Section 6.5.3**.

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Table 6.5.1-1 Summary of Process Parameter Classification and Ranges Affecting Host Cell-derived Protein Clearance

Variable	HCP Clearance Acceptable Ranges	Classification (for HCP Clearance)
AEX load pH	≥7.0	CPP
Protein load (AEX Resin)	≤300 mg protein/mL resin	CPP
Protein load (AEX Membrane)	≤10 g protein/mL membrane	CPP
Operating flow rate	≤450 cm/h	CPP
AEX load conductivity	≤15 mS/cm	CPP
Equilibration/Wash 1 buffer conductivity	≤15 mS/cm	Noncritical
Equilibration/Wash 1 buffer pH	≥7.0	Noncritical
Peak collection start	≥0.1 OD* (UV)	Noncritical
Peak collection end	≥0.1 OD* (UV)	Noncritical
Load buffering salt system	Citrate, phosphate, or tris	Noncritical
Protein concentration in load	≤50 mg/mL	Noncritical
Claimed log reduction factor (LRF) values	≥5.5 LRF for xenotropic murine leukemia virus ≥4.0 LRF for minute virus of mice	
*OD = Optical Density		

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Table 6.5.1-2 Summary of Design Parameter for AEX Step

Process Parameter	Acceptable Ranges for AEX Step	Acceptable Ranges for HCP Clearance	Parameter Classification	Supporting Data	Control Strategy
Equilibration/Wash Buffer Conductivity (mS/cm)	1.6–3.6	≤15	CPP	Multivariate study	Batch record procedure
AEX Load pH	7.2–7.8	≥7.0	CPP	Multivariate study, modular viral clearance	Batch record procedure
Protein Load (g/L resin)	50-300	≤300	CPP	Modular viral clearance study	Batch record procedure
Operating Flow Rate (cm/h)	≤450	≤450	CPP	Modular viral clearance study	Skid control
AEX Load Conductivity (mS/cm)	3.0–8.0	≤15	CPP	Modular viral clearance study	Batch record procedure
Equilibration/Wash 1 buffer pH	7.2–7.8	≥ 7.0	Noncritical	Multivariate study	Batch record procedure
Peak Collection Start A280	0.1-2 OD ascending	≥ 0.1 OD ascending	Noncritical	Multivariate study	Skid control
Peak Collection End A280	0.1-2 OD descending	≥0.1 OD descending	Noncritical	Multivariate study	Skid control

6.5.2 Reprocessing Triggers

Based on a review of the development history and the prior risk assessment, an HCP critical in-process test following the AEX step was designed and implemented to serve as the trigger for an investigation (i.e., if HCP is >100 ng/mg). The test results were available within eight (8) hours of sampling. The requirement for this additional eight-hour hold time was considered when designing the reprocessing step, as described in **Section 6.4.3** and **Sections 6.5.6-6.5.9**. The introduction of this IPC test was a permanent process change that affected all production batches and not just the reprocessed batches. The investigation procedure was defined in an SOP. If the investigation confirmed the positive result, then the reprocessing step would be initiated.

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6.5.3 Reprocessing Design

The design of the AEX reprocessing step took into consideration the issues described in **Section 6.3.1, Section 6.3.2, Section 6.5.1,** and **Section 6.5.2**. Since the step operated in flow-through mode, the AEX eluate was considered to be suitable as a feed-stream for AEX reprocessing without any adjustments. Prior development studies reviewed in **Section 6.5.1** showed that all ranges for feed-stream conditions expected for the reprocessing operation were evaluated and shown to produce drug substance with acceptable quality. In addition, studies for the reprocessing step demonstrated that the AEX column—after being stripped, cleaned, and re-equilibrated—would readily remove HCP to well below acceptable levels. All other aspects of the design of the reprocessing step, including operating conditions, buffer solution, equipment sizes, and others were similar to the design of the original AEX step.

6.5.4 Design Impact Assessment

After completing the design of the reprocessing step, the broader impact of the design on manufacturing operations was assessed. Since the AEX is a flow-through step, no adjustment to the buffer composition was required. The protein concentration was expected to be 30% lower than the original feed-stream but still within acceptable limits. The AEX load concentration is not considered a CPP.

A summary of the design impact assessment is included in **Table 6.4.1-1** in the "Other" category. The two most significant areas of concern identified were Buffer/Solution Supply and Equipment Cleaning. The maximum hold time for AEX eluate had to be increased by 48 hours to allow for completion of the HCP IPC test and to complete preparation of equipment and buffers in the event that the reprocessing trigger occurred. An evaluation of the manufacturing-scale microbial hold studies supported the longer hold time.

6.5.5 Reassess Criticality of Process Inputs

A risk analysis was also performed to assess the effect of repeating the AEX chromatography step on the criticality of process parameters in **Table 6.4.1-1.** For all parameters listed, it is assumed that the company has the ability to routinely detect an exceedance resulting in Detectability scores of 1. The risk associated with one input process parameter (hold time at pH 7.5 in **Table 6.5.5-1**) was determined to require additional development work to support longer hold times. A selection of other process inputs unrelated to operating parameters (e.g., raw materials, equipment cleaning) was also included in the FMEA assessment. (See "Other" in **Table 6.4.1-1.)** The scoring criteria and the corresponding scales are summarized in **Table 6.4.1-1** and **Table 6.5.5-1**.

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Table 6.5.5-1 AEX Reprocessing Step Process Parameter FMEA

	FMEA Scoring Process Parameter or Other Consideration	S	O	D	RPN	Comments (S, O, D)
Input AEX Process Parameters	AEX load pH	5	1	1	5	CPP, low occurrence, detectable
	Protein load (AEX resin)	5	1	1	5	CPP low occurrence, detectable
	Operating flow rate	5	1	1	5	CPP, low occurrence, detectable
	AEX load conductivity	5	1	1	5	CPP, low occurrence, detectable
	Equilibration/Wash 1 buffer conductivity	1	1	1	1	GPP, low occurrence, detectable
	Equilibration/Wash 1 buffer pH	1	1	1	1	Noncritical, low occurrence, detectable
	Peak collection start	1	1	1	1	Noncritical, low occurrence, detectable
	Peak collection end	1	1	1	1	Noncritical, low occurrence, detectable
	Load-buffering salt system	1	1	1	1	Noncritical, low occurrence, detectable
	Protein concentration in load	1	5	1	5	Noncritical, reprocessing will affect load concentration, detectable
	Hold at pH 7.5[1]	3	5	1	15	Noncritical, reprocessing will affect hold time, detectable

[1]Included from the A-Mab process cause and effect matrix (Table 4.28)

6.5.6 Scaled-Down Models

The qualified scaled-down (laboratory-scale) models used in original AEX development studies were appropriate for reprocessing development. Data demonstrating the validity of the laboratory-scale models were available, and the model was shown to be predictive of manufacturing-scale behavior.

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6.5.7 Representative Materials

Since the anion-exchange load-intermediate feed-stream from manufacturing was not available, the load for laboratory-scale models was obtained from a representative pilot-scale run that was spiked with HCP derived from the host cell line. In addition, it was possible to run the pilot-scale process under conditions that could intentionally deliver a higher-than-normal level of HCP as source material for the laboratory-scale models. All buffers and reagents were equivalent to those used in GMP manufacturing.

6.5.8 Development Studies

Based on the AEX reprocessing risk assessment in **Section 6.5.5** (**Table 6.4.1-1** and **Table 6.5.5-1**), additional development studies were performed to achieve the following:

- Expand the maximum acceptable hold-time process acceptable range (PAR) at ambient temperature
- Evaluate at laboratory-scale the effect of the reprocessing AEX column chromatography on product quality
  - Consider reprocessing the material through the step twice (i.e., double-reprocessing) as a worst-case challenge.
  - Verify that the use of reprocessed AEX eluate did not have an impact on virus removal LRF in the virus-retentive filter (subsequent processing step).
  - Evaluate the need for this viral-clearance study based on development data encompassing relevant process conditions.

6.5.9 Development Data Assessment

The data from the development studies outlined in **Section 6.5.8** were evaluated to determine whether product quality and process performance of the newly designed reprocessing step met all the comparability criteria described in **Section 6.4.3**.

6.5.9.1 Intermediate Hold-Time Study

Based on the design impact evaluation in **Section 6.5.4**, it was determined that the maximum hold time for AEX eluate had to be increased by 48 hours to allow for completion of the HCP IPC test and to complete preparation of equipment and buffers from the end of the first AEX elution. The original PAR for ambient hold time was ≤72 hours. A study was therefore performed to extend this ambient hold-time PAR to ≤120 hours. The resulting evaluation of the relevant intermediate quality attributes (identified in **Section 6.5.2**) showed that the PAR extension to ≤120 hours was acceptable. No additional CPPs were identified as a result of this study.

6.5.9.2 Reprocessing at Laboratory Scale

AEX eluate derived from the pilot run described in **Section 6.5.7** was collected and tested for all relevant intermediate quality attributes (identified in **Section 6.4.2**). This intermediate material was required to have acceptable levels of all relevant quality attributes, except for HCP, which was manipulated as described in **Section 6.5.7** to have higher than normal levels of HCP.

The intermediate material was then reprocessed twice through the qualified reduced-scale AEX step with no manipulation of the process intermediate between runs. This double-reprocessed material was used as it theoretically represents a worst-case scenario and provides stronger evidence that the reprocessing has no

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impact on product quality. The reprocessing runs were performed under target conditions each time, and samples for quality attribute testing were collected after each AEX run. The double-reprocessing of the laboratory-scale AEX eluate was performed in duplicate (four reduced-scale AEX runs in total). The test results from each run (i.e., from the original pilot run AEX eluate, from the single-reprocessed AEX eluates, and from the double-reprocessed AEX eluates) were compared. The acceptance criteria for the testing included demonstrating the reprocessed AEX eluate quality attributes were consistent with nonreprocessed material, except for the determined levels of HCP. HCP levels were required to meet drug substance release limits after the first reprocessing runs. These acceptance criteria were met. No additional CPPs were identified as a result of this study.

6.5.9.3 Virus Removal Verification

Since the reprocessed AEX eluate could impact subsequent viral removal steps, an evaluation of the performance of the reprocessing step for potential impact on the subsequent virus removal filter was performed. It was found that the minor changes to the virus filter step load pH, conductivity, and product concentration were within the originally studied virus filter step parameter ranges. Therefore, no additional virus removal studies were required for the implementation of this AEX reprocessing step.

6.5.10 Finalize Design

Based on the outcomes of laboratory-scale development, the proposed manufacturing-scale procedures were finalized. This included creation of an SOP describing reprocessing, an SOP describing the new HCP IPC test, and creation of a batch record to be used in the event of a GMP reprocessing step. No operating ranges were changed except for the maximum AEX eluate hold time at ambient temperature, which was extended from 72 to 120 hours.

6.5.11 Transfer to Manufacturing

A technology transfer project team, including all relevant stakeholders, was formed. All standard company guidelines and procedures for technology transfer were followed. As part of the technology transfer project, one intermediate pilot-scale run was executed by manufacturing personnel under draft procedures and batch records. A technology transfer report was written and approved, and all relevant staff was formally trained on the new procedures.

6.6 Conclusions

Reprocessing activities involving AEX is a feasible option during routine manufacturing. However, proper documentation aligned with GMPs is required to demonstrate the controls in place and effectiveness of the step. Among the considerations for reprocessing are:

- Prospective or concurrent validation protocol
- Clear rationale for reprocessing (e.g., reason for reprocessing)
- Proper management of the column and resin
- Proper management of the intermediate material requiring reprocessing
- Proper documentation (e.g., procedures and batch records in place)

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7.0 Appendix II: Case Study — Filtration

Filtration is among the more likely unit operations to be subjected to reprocessing; therefore, it is discussed separately in this case study. The complexity and quality impact of filtration reprocessing (referred to hereon as refiltration) tend to be low compared to other unit operations, such as the chromatography example discussed in **Section 6.0 (Appendix I)**.

The most common refiltration scenarios would include bioburden-reduction filters and virus-reduction filters, as further elaborated in **Section 7.2** and **Section 7.3**. Reprocessing of tangential flow filtration is more complex and would tend to follow a framework more similar to chromatography (as described in **Section 6.0**). Refiltration for nonproduct-containing solutions, such as media and buffer, is generally acceptable and could be desirable (e.g., due to supply constraints on certain ingredients), in which case, potential impact (such as adsorption) should also be evaluated. For most refiltration steps, additional nonroutine product and stability testing of the drug substance and drug product may not be required.

- For refiltration of an in-process bioburden-reduction filter that is used throughout the process, routine release and hold-time stability data may be adequate to support validation of reprocessing.
- Refiltration may also occur during bulk drug substance filling, where the final container was found compromised (e.g., bioprocess container (Single Use systems) found to be leaking). Reprocessing at that point may require additional validation, such as pooling of filtered and unfiltered bulk drug substance and additional bioburden-testing.
- Validation of reprocessing using a viral filter could be performed using routine product-testing, assuming the hold-time data is adequate and viral segregation is maintained.

Scenarios where product quality is compromised, such as if the drug substance or drug product exceeds a bioburden specification, if unexpected precipitation is observed in the drug substance or drug product, or if viral segregation is compromised, then reprocessing is unlikely to be an option.

The following case studies are based upon the experience of the PDA Technical Report Team members.

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7.1 Case Study 1: Proactive Drug Substance Refiltration

7.1.1 Case Study 1: Issue

The manufacturer validated the process for filling a liquid biologic drug substance into the drug substance containers (e.g., carboys, single-use bags). Prior to validating the filling process, a failure modes analysis was performed to identify potential failures that could occur during the filling process. Three potential failures were identified:

- Mechanical failures during bulk fill (e.g., structural failure such as leakage in the feed vessel, pinholes in the feed tubing, and/or single-use bags, post-use filter-test failure, failure in the filling apparatus)
- Out-of-range results of protein concentration (e.g., higher than specification, which requires further dilution with formulation buffer and refiltration) and/or spiked excipients (e.g., lower than specification polysorbate, which requires additional spiking and refiltration)
- Bioburden limits exceeded the action limit but were less than the specification

7.1.2 Case Study 1: Considerations

Up to three full-scale refiltration validation runs would be performed against a preapproved protocol capturing the failure mode, of which at least one run evaluated the failure mode of interest. Each individual refiltration protocol would be performed on a separate lot of drug substance. Samples from the validation runs would then be tested according to the specification to confirm meeting release criteria, and if necessary, placed on GMP stability for the shelf life of the drug substance. The resultant data package would provide additional confirmation that refiltration had no impact on product quality at release as well as on stability.

7.1.3 Case Study 1: Resolution

The results generated during the validation would then be filed in the BLA. The application would indicate that refiltration could only be performed once and that it must occur within a specified time period supported by the validations (e.g., validated hold stability). The application should specify that the Quality Assurance Unit must be notified and provide approval of refiltration. The refiltration procedure could be deemed as acceptable for approval as part of the license.

7.2 Case Study 2: Proactive Virus-Removal Refiltration

7.2.1 Case Study 2: Issue

Post-use integrity test failures of a normal-flow virus-retentive filter are a known failure mode affecting lot release. Refiltration of the virus-filtered pool through a new integral virus filter can be performed to ensure the validated clearance of virus; however, this reprocessing procedure has the potential to impact product quality due to extended processing times, for which the product load and/or pool stability should be considered.

7.2.2 Case Study 2: Considerations

Small-scale validation studies can be performed using a representative scaled-down model to assess the potential impact to product quality resulting from refiltration of virus-filtered pools. For these studies, samples of virus-filtration load can be obtained from different lots at manufacturing scale. Each of these

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virus-filtration loads should be refiltered at laboratory-scale through two sets of new, integral virus filters to represent worst-case. The validation studies can help to confirm that refiltration had no impact on product quality.

7.2.3 Case Study 2: Resolution

Results from the small-scale study of refiltration through a virus-retentive filter were reported in a regulatory submission. Validation of the refiltration unit operation would be completed for one full-scale batch and would include data on product yield, quality, and stability. Upon completion of the validation study and submission of the data, the refiltration procedure would become an amendment to the license.

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